

REMIX: Pooling under Forgetting for Probabilistic Intermittent-Demand Forecasting

Zong-Han Bai Po-Yen Chu

Abstract

Intermittent-demand forecasters must decide how to pool evidence across items and how much history to retain. These choices are usually specified separately, although discounting changes both an item’s reliance on a shared prior and the information available to estimate that prior. We introduce REMIX, a scalable hierarchical empirical-Bayes forecaster that applies one exponential recency operator to item- and group-level sufficient statistics and selects the discount jointly with the pooling partition by chronological validation. Forgetting has two opposing effects: it preserves prior leverage while reducing the effective sample available to resolve group heterogeneity. A two-sided credibility bound makes the trade-off explicit, and its resolution condition is estimable from the fitting window *before* any partition is fitted. Across five public panels comprising more than 19,000 series, REMIX attains the lowest mean scaled pinball loss on four, with an official Tweedie–Gaussian-process specialist ahead on Carparts; under strict walk-forward updating it improves on adaptive conformal inference by 6.2% on a long daily panel and on the strongest statistical comparator by 9.9% on a short monthly one. The margins are modest, and the diagnostic explains what bounds them: the credibility weight an item retains after discounting caps how far any shared prior can move its posterior. That bound is not what binds in practice. On the panel with the most room—a median size-block weight of 0.553, where a supplied true partition earns over two percent in controlled study—the learned partition returns -0.05% . Across all five panels, refining the pooling structure improves external accuracy on none, and the internal gain exceeds selector noise on only one. The constraint is therefore not that cross-series structure is unavailable to exploit, but that current mixture objectives do not recover a partition worth shrinking toward; we give the diagnostic that separates the two cases and show that all five public panels fall on the second side. Two evaluation artifacts recur throughout: absolute-error metrics reward the all-zero forecast on three panels, and the

standard split-conformal wrapper understates the conformal baseline by roughly 20% relative to its adaptive counterpart. The contribution is therefore a cheap, deterministic predictive law together with an ex ante test of whether cross-series pooling is worth attempting at all.

1 Introduction

Intermittent demand, in which long runs of zeros are interrupted by sporadic positive quantities, is pervasive in spare-parts logistics, long-tail commerce, and supply chains. Forecasting many such series poses two coupled statistical choices. Evidence can be pooled across items to stabilize sparse estimates, but an overly coarse pool hides real heterogeneity. Old observations can be discounted to track drift and obsolescence, but forgetting also removes information. Local methods such as Croston, SBA, and TSB [5, 23, 25] emphasize temporal adaptation while estimating each item separately. Hierarchical probabilistic models [2, 18, 20] emphasize cross-series shrinkage, usually under a fixed pooling structure or a pre-specified temporal model. Pooling and forgetting are consequently often designed as separate decisions.

Figure 1 makes both choices concrete in real intermittent demand: the relevance of an item’s own history changes over time, while the histories from which it might borrow strength are sharply heterogeneous.

They are not separate statistically. With exponential forgetting, an item’s effective history remains bounded, so a shared prior can retain substantial influence even after a long observation sequence. The same discounting, however, reduces the effective information used to estimate each group-level prior. Stronger forgetting can therefore make pooling more consequential while making fine pooling structure harder to resolve. This tension is the paper’s organizing principle: the forgetting rate sets not only a temporal horizon, but also the statistical resolution at which cross-series structure can be learned.

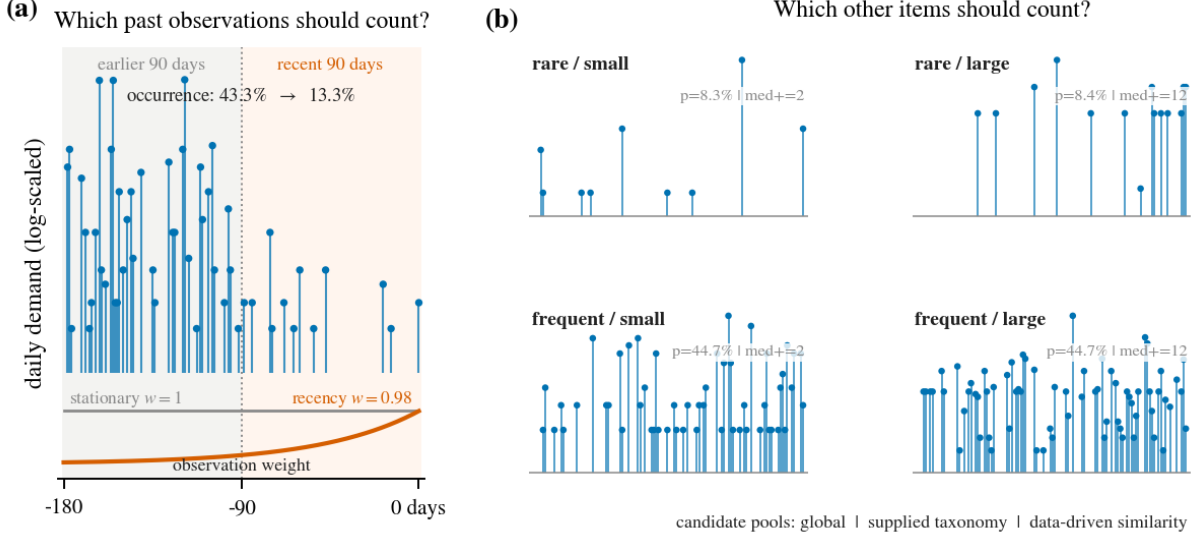


Figure 1: The two coupled choices, in real intermittent demand. **(a)** One item’s daily history: occurrence falls from 43.3% to 13.3% between the earlier and the recent ninety days, so a stationary weighting ($w = 1$) and a recency weighting ($w = 0.98$) summarize the same history differently—which past observations should count? **(b)** Four items from the same panel, spanning rare and frequent occurrence at small and large sizes: any pooled prior must decide which of these heterogeneous neighbors an item should borrow strength from—which other items should count? Candidate pools are a global pool, a supplied taxonomy, and data-driven similarity.

We operationalize this principle with REMIX,¹ a recency-weighted hierarchical empirical-Bayes forecaster for intermittent demand. The model uses an occurrence–size decomposition with Beta–Binomial occurrence and log-normal size blocks, represented by per-item sufficient statistics. The same exponential weights are applied to the statistics used for the item posterior and for group-level hyperparameter estimation. A small candidate family crosses discount factors with a global pool, the classical ADI/CV² taxonomy, and BIC-selected mixture partitions; chronological validation selects the pair before a full-window refit. For a fixed candidate family, estimation is deterministic and linear in the number of observations, with low-dimensional group optimization and mixture search.

The theory predicts an asymmetric empirical pattern. Forgetting should be a first-order forecasting decision, whereas refining an already pooled model can help only when the discounted data still resolve the proposed groups. Because the resolution condition depends on quantities available from the fitting window, it can be checked panel by panel before any

partition is fitted. The first half holds: all five panels select a discount below one, and on the panel where both effects are measurable forgetting is worth an order of magnitude more than refinement. The second half fails in a more informative way than we expected. Refining the pooling structure improves external accuracy on none of the five panels, and the diagnostic shows this is not because the panels lack room. Three of them retain substantial credibility leverage, and one—Online Retail—sits at exactly the level where a supplied true partition earns more than two percent in controlled study, while its learned partition returns -0.05% . What binds is therefore not the resolution limit but learnability: the structure would pay if it were correct, and the mixture objective does not find it. Separating those two explanations is what the diagnostic is for, and it is the distinction between a resolution limit and an absence of useful cross-series structure.

The resulting empirical position is deliberately narrow. Across the five panels REMIX attains the lowest mean scaled pinball loss on four, with margins over the strongest comparator ranging from roughly one percent to ten; an official Tweedie–Gaussian-process implementation is better on Carparts and throughout that panel’s far tail. Under strict walk-forward updating the margin is 6.2% over adaptive

¹The name reflects the construction rather than an acronym: recency-weighted evidence and cross-series pooling are *remixed* inside one empirical-Bayes family instead of being specified as separate decisions.

conformal inference on Online Retail and 9.9% over the strongest statistical comparator on Carparts, and it is concentrated in the upper quantiles: below the median, several methods on these panels coincide exactly with the all-zero forecast. We report this structure rather than an aggregate win because it is what the mechanism predicts, and because two widely used evaluation conventions actively obscure it.

Contributions. (1) **Joint adaptation.** We introduce a scalable empirical-Bayes model that applies one recency operator consistently to item- and group-level sufficient statistics, then selects the discount jointly with the pooling structure by leakage-free chronological validation. The resulting predictive law is closed form, bit-reproducible, and costs milliseconds per series on a CPU. (2) **An estimable resolution condition.** We show that discounting has two linked effects: it keeps prior leverage active but reduces the information with which group heterogeneity can be estimated. A two-sided credibility bound gives the collapse probability of the group variance component, and its resolution corollary converts that bound into a statistic computable from the fitting window. The statistic bounds how much any partition could be worth on a given panel before one is fitted, and separates a panel with no room from a panel whose room we fail to use. Connections to TSB, drift tracking, classical credibility, and finite-class selection are collected in the appendix. (3) **Mechanism-focused evidence, including a negative result.** Five real-data panels, two strict walk-forward evaluations, and a known-partition simulation separate gains from forgetting, gains from pooling refinement, and distortions caused by zero-favoring point metrics. Comparators include the official TweedieGP implementation, adaptive conformal inference, a published hierarchical state-space model, and a zero-shot foundation model. Learned pooling improves external accuracy on none of the five panels, and we show this is not attributable to a lack of leverage: on the panel with the most room the learned partition still loses. We also document two evaluation failures that the intermittent-demand literature currently inherits—absolute-error metrics reward the all-zero forecast, and static split-conformal wrappers understate what conformal prediction can do here by about 20%.

2 Related Work

Intermittent-demand forecasting. Croston’s decomposition [5], its SBA correction [23], and TSB [25] estimate occurrence and positive size locally; TSB additionally lets the occurrence estimate decay under obsolescence. ADIDA [17] and IMAPA [12] instead reduce intermittency through temporal aggregation, while likelihood-based alternatives include inventory count models [21], iETS [22], Tweedie Gaussian processes [6], and deep renewal processes [27]. Conformal wrappers turn point forecasters into predictive intervals, and adaptive conformal inference updates miscoverage levels under distribution shift [7]; we evaluate both the static and the adaptive form, because the difference between them is larger than most method differences reported on these panels. These approaches provide important forms of temporal adaptation, but do not jointly learn how recency-weighted evidence should be shared across items. Hierarchical models do share evidence: Chapados [2] and Seeger et al. [20] pool related count series at scale, and Pitkin et al. [18] develops a hierarchical hurdle model for retail. Their pooling structure or temporal evolution is generally specified in advance. REMIX instead crosses recency and pooling choices within one occurrence–size empirical-Bayes model, and asks when that crossing can pay.

Learning cross-series structure. The continuum between local and global forecasting is formalized by Montero-Manso and Hyndman [16]. More recent work learns richer forms of cross-series specialization. For hierarchical forecasting, Cini et al. [4] learns a graph-pooled aggregation hierarchy aligned with the forecasting objective. Moirai-MoE [14] replaces human-defined frequency groups with token-level sparse-expert routing in a time-series foundation model. These approaches motivate data-adaptive structure, but address different objects: aggregation coherence or latent expert specialization. Our groups are explicit exchangeability pools for a shared prior. That restriction is what makes their statistical resolution analyzable: the quantity governing whether a pool can be estimated is the same credibility weight that governs whether it can matter, so a single diagnostic covers both. It also permits a direct comparison among global, fixed-rule demand-pattern, and learned mixture pools under one predictive law.

Discounting, credibility, and selection. Exponential discounting is classical in Bayesian dynamic models [26], and power priors down-weight historical likelihoods [9]. Likewise, $\lambda_i = n_i/(n_i + \phi_g)$ is the classical actuarial credibility form [1]; Proposition 6 is therefore a restatement, not a claimed

contribution. The distinction in REMIX is that one recency operator updates both item statistics and the empirical-Bayes statistics defining the shared prior, and its strength is selected jointly with the pool. Proposition 2 then identifies the resulting two-sided resolution limit: forgetting preserves prior leverage but can erase the information needed to distinguish fine pools. Chronological validation selects among the finite candidate family. Because zero-heavy point losses can favor degenerate forecasts, selection uses pinball loss, a proper scoring rule [8], with the high quantiles relevant to inventory decisions [11].

3 Remix: Recency-Weighted Hierarchical EB

REMIX combines two candidate choices inside one occurrence-size model: a recency discount w determines which observations remain informative, and a partition g determines which items share an empirical-Bayes prior. The key construction is that the *same* discounted sufficient statistics determine both the item posterior and the group prior. We first define that representation, then the candidate partitions and their leakage-safe joint selection.

3.1 Occurrence-size model

For item i , let $z_{i,t} = \mathbf{1}\{y_{i,t} > 0\}$ and define $\ell_{i,t} = \log y_{i,t}$ when $z_{i,t} = 1$. Conditional on membership $g_i = g(i)$, the hurdle model is

$$\begin{aligned} \pi_i &\sim \text{Beta}(\alpha_{g_i}, \beta_{g_i}), & z_{i,t} \mid \pi_i &\sim \text{Bern}(\pi_i), \\ \mu_i &\sim \mathcal{N}(\mu_{0,g_i}, \tau_{g_i}^2), \\ \sigma_i^2 &\sim \text{ScaleInv-}\chi^2(\nu, s_{0,g_i}^2), \\ \ell_{i,t} \mid z_{i,t}=1, \mu_i, \sigma_i^2 &\sim \mathcal{N}(\mu_i, \sigma_i^2). \end{aligned} \quad (1)$$

Here $s_{0,g}^2$ is parameterized so that the prior mean of σ_i^2 is the group estimate σ_g^2 ; ν is fixed rather than selected. Group hyperparameters are estimated by empirical Bayes: Beta-Binomial marginal likelihood for occurrence, and an intercept-only random-effects fit plus conjugate variance shrinkage for positive log-size. Appendix D gives the numerical objectives and fallback rules.

Without forgetting, all inference depends on the four statistics

$$S_i(1) = (n_i, c_i, u_i, v_i) = \sum_{t=1}^{T_i} (1, z_{i,t}, z_{i,t}\ell_{i,t}, z_{i,t}\ell_{i,t}^2),$$

where the last two terms are zero when $z_{i,t} = 0$. The occurrence posterior mean and the positive-size

posterior location take parallel credibility forms,

$$\begin{aligned} \hat{\pi}_i &= \lambda_i^{(o)} \bar{p}_i + (1 - \lambda_i^{(o)}) \mu_{\pi, g_i}, & \lambda_i^{(o)} &= \frac{n_i}{n_i + \phi_{g_i}}, \\ \hat{\mu}_i &= \lambda_i^{(+)} \bar{\ell}_i + (1 - \lambda_i^{(+)}) \mu_{0, g_i}, & \lambda_i^{(+)} &= \frac{n_i^+}{n_i^+ + \kappa_{i, g_i}}, \end{aligned} \quad (2)$$

with $\phi_g = \alpha_g + \beta_g$ and $\kappa_{i, g} = \hat{\sigma}_i^2 / \tau_g^2$. Thus a partition matters only through the prior target and strength supplied to its members.

Two variance ratios appear in what follows. Estimation uses the item-level $\kappa_{i, g} = \hat{\sigma}_i^2 / \tau_g^2$ in (2); the analysis in Section 4.2 works with the group-level surrogate $\kappa_g = \sigma_g^2 / \tau_g^2$, which replaces each item's process variance by its group estimate. The two coincide when the group is homogeneous in σ_i^2 , and the surrogate is what the collapse event of Proposition 2 is stated in terms of.

The plug-in marginal predictive distribution is zero-inflated log-normal:

$$\begin{aligned} \Pr(Y_i = 0 \mid \mathcal{D}) &= 1 - \hat{\pi}_i, \\ \log Y_i \mid (Y_i > 0, \mathcal{D}) &\sim \mathcal{N}(\hat{\mu}_i, \hat{\sigma}_i^2 + V_i), \end{aligned} \quad (3)$$

where V_i is posterior uncertainty in μ_i . Its quantiles are analytic. For point evaluation we retain the conditional plug-in mean $\hat{y}_i = \hat{\pi}_i \exp(\hat{\mu}_i + \hat{\sigma}_i^2/2)$; it is distinct from the mean of (3), which also integrates V_i .

3.2 Shared recency weighting

For $w \in (0, 1]$, replace $S_i(1)$ by

$$S_i(w) = \sum_{t=1}^{T_i} w^{T_i-t} (1, z_{i,t}, z_{i,t}\ell_{i,t}, z_{i,t}\ell_{i,t}^2). \quad (4)$$

These fractional counts are valid weighted-likelihood statistics; the log-Gamma form of the occurrence marginal and the Normal size objective extend directly to them. Crucially, $S_i(w)$ is used twice:

$$\begin{aligned} \hat{\eta}_g(w) &= \arg \max_{\eta_g} \sum_{i: g_i=g} \log m\{S_i(w) \mid \eta_g\}, \\ q_i^{(w)}(\theta_i) &= q_i(\theta_i \mid S_i(w), \hat{\eta}_{g_i}(w)), \end{aligned} \quad (5)$$

where η_g collects the group hyperparameters and q_i is the item posterior. The group prior therefore adapts to recent cross-sectional evidence rather than remaining a stationary prior applied to discounted item data. Setting $w = 1$ recovers the stationary estimator exactly.

Table 1: Pooling candidates used by REMIX, ordered coarse to fine. Every data-dependent label is computed from the selector’s fitting head, never its validation tail.

Candidate	Shared-prior groups
Global	All items share one prior.
Taxonomy	Four ADI/CV ² demand-pattern classes (smooth, intermittent, erratic, lumpy) using the classical thresholds 1.32 and 0.49 [24].
Learned	Hard assignments from a mixture of occurrence-size prior components; $K \in \{1, \dots, 6\}$ is selected by BIC.

For occurrence, (2) becomes

$$\begin{aligned}\hat{\pi}_i(w) &= \lambda_i(w)\bar{p}_i(w) + \{1 - \lambda_i(w)\}\mu_{\pi, g_i}(w), \\ \lambda_i(w) &= \frac{n_i(w)}{n_i(w) + \phi_{g_i}(w)}.\end{aligned}\tag{6}$$

The analogous positive-size update uses discounted positive mass $n_i^+(w)$ and the group variance ratio $\kappa_{i, g}(w)$.

3.3 Candidate pooling structures

Let $\mathcal{R}$ contain three ways to construct $g(\cdot)$ from the fitting data. Table 1 orders them from coarse to fine: one prior for the panel, one prior per fixed demand-pattern class, and one prior per learned component. The ladder is deliberate. The question this paper asks is not which rung is best in general—that depends on the panel—but whether the discounted data support moving up it at all. Section 4.2 gives the condition under which they do, and Section 5.2 reports what happens when they do not.

For the learned candidate, component k carries $\eta_k = (\alpha_k, \beta_k, \mu_{0, k}, \tau_k^2, \sigma_k^2)$. Its item marginal is the product of Beta-Binomial occurrence evidence and Normal random-effects size evidence. EM has an analytic responsibility update; its M-step maximizes responsibility-weighted EB objectives. A deterministic quantile initialization makes repeated fits seed-free. Mixture labels are built once from the fitting head and shared across the discount grid, with $K \in \{1, \dots, 6\}$ selected by BIC at that fit. Re-learning the partition separately at every discount is a natural variant—changing w changes both the item evidence and the components it can support—but we do not evaluate it here. The selected responsibilities are converted to hard labels so all three candidates share the same downstream forecaster.

3.4 Leakage-safe joint selection

Each training series is split chronologically into a fitting head $\mathcal{H}_i$ (first 80%) and validation tail $\mathcal{V}_i$ (last 20%). For every $(r, w) \in \mathcal{R} \times \mathcal{W}$, with

$$\mathcal{W} = \{1, .999, .997, .995, .99, .98, .95, .90\},$$

the partition builder for r sees only $\mathcal{H} = \cup_i \mathcal{H}_i$. The resulting labels, discounted EB prior, and item posteriors are all fit on the same head. The candidate is scored by

$$\begin{aligned}\hat{R}(r, w) &= \frac{1}{N_V} \sum_{i \in \mathcal{I}_V} \frac{1}{|\mathcal{V}_i| |\mathcal{Q}|} \\ &\times \sum_{t \in \mathcal{V}_i} \sum_{q \in \mathcal{Q}} \frac{\rho_q(y_{i, t}, \hat{Q}_{i, t}^{(r, w)}(q))}{a_i},\end{aligned}\tag{7}$$

where $\mathcal{Q} = \{.5, .75, .9\}$, ρ_q is pinball loss, and a_i is mean absolute demand on the fitting head (with a panel-level fallback for an all-zero head); $N_V = |\mathcal{I}_V|$ is the number of scored items. Times and quantiles are averaged within item before items receive uniform weight. This matches the panel-level goal and prevents long series or high-volume items from dominating selection.

The selected pair is $(\hat{r}, \hat{w}) = \arg \min_{r, w} \hat{R}(r, w)$. Its partition is then *relearned* on the complete training window at $\hat{w}$, and the EB model is refit once before external evaluation. Thus the validation tail chooses a construction rule and discount, not a permanent set of labels; external targets are never used. There are 24 outer forecast candidates.

Resolution diagnostics. Fitting each candidate already produces everything the resolution analysis of Section 4.2 needs, so the selector records it rather than recomputing it later. For every (r, w) and every resulting group g we store, from the fitting head only, the median occurrence and positive-size credibility weights $\lambda_i^{(o)}$ and $\lambda_i^{(+)}$, the share of items with $\lambda_i^{(+)} < 0.1$, the group sizes m_g , and the resolution statistic

$$\hat{z}_g = \frac{S_g^2 - \bar{s}_g}{\sqrt{2/(m_g - 1) S_g^2}},$$

whose nonpositive values coincide with the collapse event $\hat{\tau}_g^2 = 0$ of Proposition 2. These quantities cost nothing beyond the fit that produced them, they are computed before any external observation is seen, and they answer a question the validation score cannot: not which candidate scores best on the tail, but whether the discounted data contain enough information to estimate the candidate’s group

structure at all. Section 5.2 uses them to separate the two. The full selection surface and the accompanying diagnostics are released with every run (`remix_selection_diagnostics.csv`).

Appendix D gives split edge cases and pseudocode-level implementation details.

3.5 Forecasts and computation

Selection uses the analytic plug-in quantiles in (3). The reported probabilistic forecasts also average quantile estimates across $B = 20$ cross-item bootstrap refits of the group hyperparameters. We treat this explicitly as a quantile ensemble; the average of component quantiles is not asserted to be the quantile of a single posterior mixture.

For each w , one pass computes $S_i(w)$ in $O(\sum_i T_i)$ time. Each EB fit requires only low-dimensional deterministic optimization per group; mixture EM is $O(NK)$ per iteration. Prediction is $O(N)$ for points and $O(NQ)$ for Q quantiles.

Two properties of this cost profile matter for the comparisons that follow. First, the whole pipeline is CPU-only and closed form apart from the mixture search: fitting and predicting a panel costs on the order of milliseconds per series, so the full 24-candidate joint selection completes in minutes on a laptop-class machine and no GPU, tuning budget, or convergence diagnostic is required. Second, estimation is exact rather than approximate at every stage, so repeated end-to-end runs are bit-identical in the released test suite. Neither property is an accuracy claim; both are the terms on which the accuracy results in Section 5 should be read, since several of the strongest comparators there cost two to three orders of magnitude more per series and none is reproducible to the bit. Measured wall-clock figures for every method are in Appendix D.

4 Resolution under Forgetting

The shared operator in (5) creates two opposing effects. A bounded item history prevents the group prior from vanishing, but the same bound limits the information used to estimate fine group structure. This section formalizes that interaction. Supporting results connecting the model to TSB, drift tracking, finite-class selection, and classical credibility are collected in Appendix B.

4.1 Persistent prior leverage

Fix an item and suppress its subscripts. Because $n(w) = \sum_{t=1}^T w^{T-t} \leq (1-w)^{-1}$ for $w < 1$, the

occurrence credibility weight cannot converge to one.

Proposition 1 (Forgetting keeps pooling relevant). For every horizon T , $\lambda(w) \leq \{1 + (1-w)\phi\}^{-1} < 1$ when $w < 1$, whereas $\lambda(1) \rightarrow 1$ as $T \rightarrow \infty$. Hence the pooled occurrence prior retains positive weight indefinitely under forgetting.

The statement is about leverage, not benefit: it does not guarantee that a particular prior is a good shrinkage target. It says that the choice of target can continue to affect forecasts even for old series.

4.2 Variance-component resolution

The converse problem appears in the positive-size block. Let item i in group g have discounted positive mass n_i^+ , sampling variance $s_i = \sigma_g^2/n_i^+$ for its log-size mean, and credibility $\lambda_i^{(+)} = n_i^+/(n_i^+ + \kappa_g)$ with $\kappa_g = \sigma_g^2/\tau_g^2$. The group variance is estimated by the truncated moment/REML form $\hat{\tau}_g^2 = \max(0, S_g^2 - \bar{s}_g)$, where S_g^2 is the sample variance of the m_g item means and $\bar{s}_g = m_g^{-1} \sum_{i \in g} s_i$.

Proposition 2 (Two-sided effect of forgetting). (i) For $w < 1$, $\lambda_i^{(+)} \leq \{1 + (1-w)\kappa_g\}^{-1} < 1$. (ii) For independent item means with possibly unequal s_i ,

$$\mathbb{E}[S_g^2] = \tau_g^2 + \bar{s}_g$$

holds exactly. Under the Gaussian moment approximation,

$$\Pr(\hat{\tau}_g^2 = 0) \simeq \Phi\left(-\sqrt{\frac{m_g - 1}{2}} \rho_g\right), \quad \rho_g = \frac{\tau_g^2}{\tau_g^2 + \bar{s}_g}. \quad (8)$$

Since $n_i^+ \leq (1-w)^{-1}$, $\bar{s}_g \geq (1-w)\sigma_g^2$ and therefore

$$\rho_g \leq \frac{\tau_g^2}{\tau_g^2 + (1-w)\sigma_g^2}.$$

On the collapse event the plug-in estimate has $\hat{\tau}_g^2 = 0$ and $\lambda_i^{(+)} = 0$: the item mean is replaced by its estimated group mean.

The population ratio ρ_g is not observed. For diagnostics we use

$$\hat{z}_g = \frac{S_g^2 - \bar{s}_g}{\sqrt{2/(m_g - 1) S_g^2}},$$

whose nonpositive values coincide with the plug-in collapse event.

Corollary 1 (Resolution condition). For this estimator and window, stable estimation of a group variance requires, up to a fixed probability threshold,

$$\rho_g \sqrt{m_g - 1} \gtrsim 1, \quad \text{or equivalently} \quad \tau_g^2 \gtrsim \frac{\bar{s}_g}{\sqrt{m_g - 1}}$$

with $\bar{s}_g \geq (1 - w)\sigma_g^2$.

For similarity-based refinements, finer groups typically reduce both m_g and the within-group spread τ_g^2 , while stronger forgetting raises the sampling-variance floor. All three changes move the estimator toward collapse. This is an estimator- and window-specific resolution condition, not an information-theoretic lower bound, and it does not say which candidate wins below the threshold.

Figure 2 summarizes why pooling can become more consequential at exactly the point where a fine partition becomes harder to estimate. Section 5.4 tests both sides with known group labels. Proofs and the accuracy limits of the approximation are given in Appendix C and Section 6.

5 Experiments

5.1 Experimental design

Datasets. We evaluate on two daily panels—**Online Retail** [3] (3,649 series) and a seed-controlled 5,000-series **M5** sample [15]—and the monthly intermittent suite of Damato et al. [6]: **Auto** (3,000 series), **Carparts** (2,509), and **RAF** (5,000). Together they contain more than 19,000 series and span long daily histories and short, highly sparse monthly panels. Dataset construction and holdout lengths are in Appendix E.14. Both the probabilistic and the point M5 protocols use the first two thirds of each series for initialization.

Comparators. The classical set comprises Croston, SBA, TSB, ADIDA, and IMAPA; stronger statistical comparators are AutoARIMA, AutoTheta, a per-series autoregressive Tweedie GLM, iETS, and the official TweedieGP implementation on the three monthly panels. For probabilistic evaluation we use native distributions where available and chronological split-conformal wrappers (CP-*) for the point forecasters. The Online Retail walk-forward study additionally includes adaptive conformal inference (ACI) applied to ADIDA and TSB. A 25-candidate tuned TSB control receives the same internal chronological selection budget as REMIX. DeepState and Chronos-Bolt comparisons are confined to Appendix E.11 because they use subset or proxy protocols. Implementation details and complete settings are in Appendix D.

Protocols and metrics. The primary fixed-origin experiment fits once on the initialization window and scores the full evaluation span. A strict walk-forward experiment then reveals and updates one block at a time: seven-day blocks on Online Retail and one-month blocks on Carparts. Classical and conformal methods refit each block except Online Retail AutoARIMA, AutoTheta, and iETS, which refit every four blocks; REMIX updates discounted sufficient statistics each block. The primary metric is scaled pinball loss (SPL; lower is better), averaged over $q \in \{0.1, 0.25, 0.5, 0.75, 0.9\}$; we separately report q90 and the far tail because inventory decisions are asymmetric. All quantiles are zero-truncated and monotonized, and REMIX is reported without post-hoc calibration. Every model in Table 2 is scored by one implementation from its stored quantile predictions, so no cell mixes scoring versions. Fixed-origin, walk-forward, and subset foundation model results are kept separate because they answer different questions. MAE and RMSSE are secondary diagnostics rather than the headline criteria because zero forecasts can minimize absolute error on intermittent panels. Full point, interval, significance, and efficiency results are in Appendix E.

5.2 Probabilistic forecasting across panels

Table 2 sets the empirical position, and it is a narrow one. REMIX attains the lowest mean SPL on four of the five panels; the official TweedieGP implementation wins Carparts (0.3214 versus 0.3229). The margins over the strongest comparator vary by an order of magnitude across panels: 10.1% on Online Retail, 3.0% on RAF, 1.6% on Auto, and 1.2% on M5. Across three independent 5,000-series M5 samples the selected configuration is identical—(mixture, $w=0.95$, $K=6$)—and REMIX has mean SPL 1.7501 ± 0.084 against 1.7547 ± 0.065 for CP-Croston, its closest comparator: ahead on two samples, behind by 1.0% on the third, and ahead on average by 0.3%, well within one standard deviation. We report the spread rather than an aggregate win because it is the quantity Section 5.5 explains. The explanation is not that these panels leave no room for a hierarchical construction to work; on three of them the credibility weights say otherwise. It is that the room which exists is not converted by a learned partition, which is why the achievable margin is set by the shared predictive law rather than by the pooling structure.

The degenerate row exposes what the central quantiles cannot see. On RAF, the all-zero forecast is

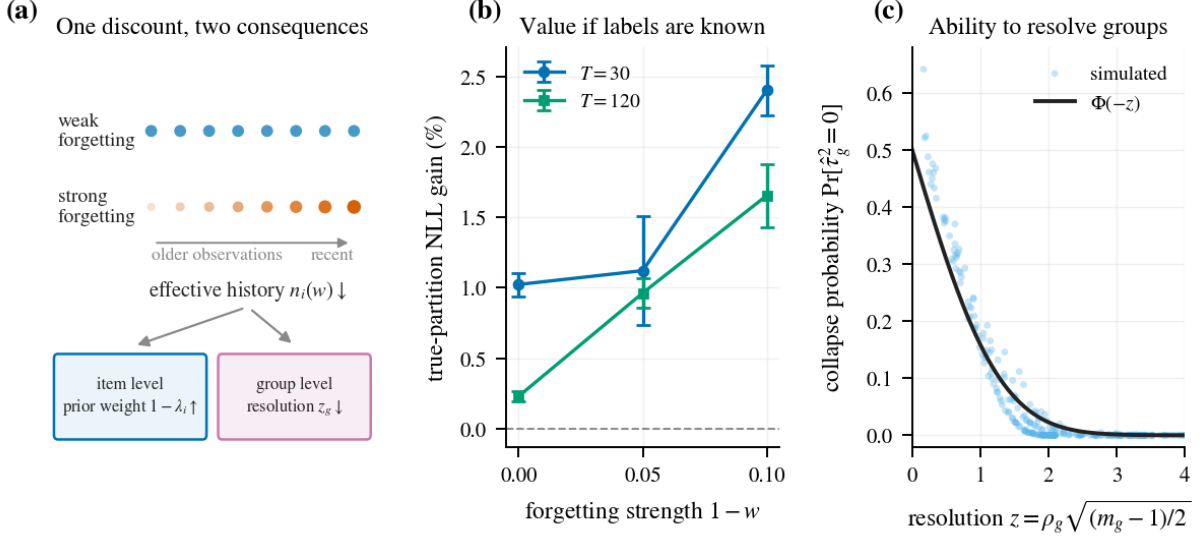


Figure 2: **One discount creates two opposing effects.** (a) The same recency operator reduces item effective history, which increases prior leverage, and group effective information, which reduces structural resolution. (b) In the controlled study, a supplied true partition becomes more valuable as forgetting strengthens. (c) When the partition must be estimated, the probability that the group variance collapses rises as the resolution statistic falls; the curve is the approximation in (8).

marginally better through q90 because the true quantiles of most series are zero. At $q \in \{0.95, 0.975, 0.99\}$ it falls behind every intensity-bearing method, and intensity-bearing models become distinguishable from one another (Appendix E.12). TweedieGP is strongest throughout the Carparts far tail and remains competitive on RAF. The probabilistic claim is therefore not that one distribution wins every quantile, but that REMIX is the most consistent aggregate performer across panels and remains informative in the region where a zero forecast ceases to be useful.

Point forecasts tell a deliberately weaker story. REMIX is best on six of ten MAE/RMSSE cells and top two on seven, but the zero forecast has the lowest MAE on three panels and classical methods lead several remaining cells. We report the complete point table and horizon decomposition in Appendix E.1: they establish competitiveness and rule out a long-horizon averaging artifact, but do not support uniform point-forecast superiority.

5.3 Strict walk-forward forecasting

Table 3 is the deployment-facing test. On Online Retail REMIX improves mean SPL by 6.2% over ACI-ADIDA and q90 by 5.6% over AutoTheta, the strongest q90 comparator. On Carparts it improves mean SPL by 9.9% over CP-IMAPA and wins all five

quantiles.

Three features of this table deserve more attention than the margins. First, *the choice of conformal construction matters more here than the choice of forecaster*: making the wrapper adaptive [7] improves ADIDA’s mean SPL from 1.1576 to 0.9108, a 21% gain that is more than three times the remaining gap to REMIX. Static split-conformal wrappers are the standard probabilistic baseline in this literature; on this evidence they understate what conformal prediction can do on intermittent panels, and comparisons drawn against them—including earlier versions of our own—are correspondingly generous. Second, *coverage and sharpness separate*. ACI-ADIDA attains higher positive-demand coverage (0.646 versus 0.621) with intervals almost twice as wide (19.67 versus 10.31); neither method dominates, and the SPL ordering reflects sharpness rather than calibration. Third, *the central quantiles carry little information on the sparse panel*: on Carparts the q10 and q25 entries of REMIX equal the all-zero forecast exactly and q50 sits marginally below it, so essentially the entire mean-SPL difference is generated above the median. We return to this in Appendix E.12, where the far-tail levels are the only ones that separate intensity-bearing models on the sparsest panels.

The configuration is selected once at the origin and then updated online; repeated online re-selection

Table 2: **Probabilistic forecasting is the primary comparison.** Scaled pinball loss (SPL; lower is better) is reported as the mean over $q \in \{0.1, 0.25, 0.5, 0.75, 0.9\}$ and at q90. Bold and underline mark the best and second-best non-degenerate methods. M5 reports seed 42 for direct comparability; three-seed robustness is summarized in the text. REMIX uses its raw predictive law, with no post-hoc calibration. TweedieGP is the official implementation and is evaluated on the three monthly panels for which the authors released it. The all-zero row is a metric diagnostic and is excluded from ranking.

Model	Online Retail		M5		Auto		Carparts		RAF	
	Mean	q90	Mean	q90	Mean	q90	Mean	q90	Mean	q90
AutoARIMA	1.0069	1.6297	1.7919	2.4148	0.3189	0.2659	0.4084	0.4897	0.4020	0.5910
AutoTheta	1.0055	<u>1.6056</u>	1.7954	2.3123	0.3403	0.2760	0.3925	<u>0.4697</u>	0.4104	0.5956
CP-Croston	1.5502	2.7686	<u>1.6902</u>	2.1551	0.3260	0.3105	0.4012	0.5433	0.3322	0.4927
CP-SBA	1.5492	2.8601	1.6939	<u>2.1736</u>	0.3256	0.3153	0.3982	0.5418	0.3290	0.4920
CP-TSB	1.1744	2.6746	1.8794	2.3148	0.3459	0.3150	0.3783	0.4842	0.3814	0.4986
CP-ADIDA	1.1239	2.5409	1.7150	2.1949	0.3275	0.3082	0.3571	0.4982	0.3097	0.4905
CP-IMAPA	1.1280	2.5506	1.7190	2.1904	0.3279	0.3084	0.3561	0.4908	0.3116	0.4907
Tweedie-GLM	1.1477	1.7458	4.0693	5.2620	0.3221	0.2783	0.4776	0.6897	0.3121	0.5312
iETS	<u>0.9846</u>	1.7544	2.0282	2.9772	0.3448	0.3252	0.4561	0.5646	0.3494	<u>0.4900</u>
TweedieGP	—	—	—	—	<u>0.2985</u>	0.2598	0.3214	0.4600	<u>0.2763</u>	0.5219
Zero (diagnostic)	0.9161	1.6490	1.9590	3.5262	0.5612	1.0102	0.3736	0.6725	0.2669	0.4805
REMIX (ours)	0.8849	1.5268	1.6707	2.4758	0.2937	<u>0.2599</u>	<u>0.3229</u>	0.4817	0.2680	0.4856

Table 3: **Strict probabilistic walk-forward evaluation.** Online Retail uses seven-day blocks and Carparts uses monthly blocks. Mean and q90 are scaled pinball losses; Cov.⁺ is 80% interval coverage conditional on positive demand and AIW is average interval width. Dashes denote an unevaluated panel. AutoTheta refits every four Online Retail blocks; the classical wrappers, ACI, and all Carparts baselines refit each block. REMIX updates sufficient statistics each block. Bold and underline rank non-degenerate methods by SPL.

Model	Online Retail				Carparts			
	Mean	q90	Cov. ⁺	AIW	Mean	q90	Cov. ⁺	AIW
AutoTheta	1.0586	<u>1.4835</u>	0.765	16.43	0.4056	<u>0.4743</u>	0.678	1.80
CP-TSB	1.3733	2.8422	0.468	5.71	0.3836	0.4758	0.551	1.27
CP-ADIDA	1.1576	2.5143	0.473	5.47	0.3509	0.4880	0.593	1.28
CP-IMAPA	1.1773	2.5224	0.471	5.47	<u>0.3492</u>	0.4789	0.581	1.27
ACI-ADIDA	<u>0.9108</u>	1.6681	0.646	19.67	—	—	—	—
Zero (diagnostic)	0.9161	1.6490	0.000	0.00	0.3736	0.6725	0.000	0.00
REMIX	0.8540	1.4002	0.621	10.31	0.3145	0.4578	0.669	1.41

is outside the present experiment.

5.4 Controlled test of the resolution mechanism

Real data never reveal the true pooling partition, so on their own they cannot distinguish a partition that is not worth having from one we failed to find. Both appear as a small number. We therefore construct panels where the partition is correct by definition, and use them as a reference against which the learned partitions of Section 5.5 are measured: what a good partition is worth at a given level of credibility leverage, rather than what our estimator obtains there.

We simulate from the hierarchical hurdle model

of (1) with group centers arranged symmetrically around a fixed grand mean, so that increasing separation does not change the panel scale. Each cell compares a single pool against the known true partition under the same predictive representation, with nine paired replicates, and sweeps the two quantities that Section 4.2 identifies as controlling credibility: the discount w and the series length T . The resulting grid spans the leverage range the real panels occupy, from the near-fully-shrunk regime of Carparts to the item-dominated regime of Auto and RAF.

Table 4 gives the result. The true partition improves NLL and log-size MSE in every cell, and pinball loss whenever forgetting is strong; in a separate length sweep its pinball advantage decays from 1.53%

Table 4: Advantage of the *true* partition over a single pool on simulated panels (mean percentage improvement over nine paired replicates; standard error in parentheses). Positive favors the partition. T is series length.

T	w	NLL	log-size	MSE	pinball	MAE
30	0.90	2.40 (0.18)	17.15 (2.39)	2.25 (0.61)	-1.72 (0.75)	0.17
30	0.95	1.12 (0.39)	9.00 (2.45)	-0.44 (0.58)	-0.17 (0.51)	0.59
30	1.00	1.02 (0.08)	8.37 (1.53)	-0.69 (0.32)	-1.95 (0.50)	0.59
120	0.90	1.65 (0.22)	18.36 (1.56)	1.57 (0.64)	-3.34 (0.47)	0.59
120	0.95	0.96 (0.11)	3.65 (0.77)	0.23 (0.28)	-1.32 (0.53)	0.59
120	1.00	0.23 (0.04)	1.13 (0.31)	-0.32 (0.13)	-0.80 (0.11)	0.59

at $T = 12$ to 0.01% at $T = 400$. Both patterns follow the credibility mechanism directly. Discounting or short histories reduce an item’s own weight $\lambda_i^{(+)}$ and open room for a prior to act; long undiscounted histories drive $\lambda_i^{(+)} \rightarrow 1$ and make every partition irrelevant. Stronger forgetting simultaneously makes group variance components harder to estimate, as Proposition 2 predicts, which is why the gains in the $w = 0.90$ rows are bought against a noisier $\hat{\tau}_g^2$ and do not continue to grow without limit.

MAE moves the other way in all six cells. A known-correct partition, on data generated from the model it describes, is reported as harmful by the metric most commonly used in this literature. We take this as the cleanest available statement of why Section 5.2 treats absolute-error criteria as diagnostics rather than as the headline.

What these panels are for. Plotted against median $\lambda^{(+)}$, the cells of Table 4 and the length sweep trace the grey curves of Figure 3: the return to *correct* structure as a function of leverage. Because the partition is supplied rather than estimated, that return is an upper envelope—no procedure that must also learn the grouping can exceed it, and any procedure that falls below it is losing something to estimation rather than to a lack of structure. This is the comparison Section 5.5 draws, and it is the reason the controlled study is necessary rather than decorative: the real panels supply the leverage coordinates, and only a known-partition experiment can supply the ordinate they should be judged against.

Two boundaries apply. The experiment isolates the value of correct structure and says nothing about whether the mixture recovers it—indeed Section 5.5 shows it does not, and the gap between the two is precisely what the comparison measures. And it simulates from the model that REMIX fits, so the mechanism is tested under correct specification only; a

misspecified size law would shift the envelope, though it is hard to see how it would raise it. Full diagnostics, including the growth of $\text{RMSE}(\hat{\tau}_g^2)$ as w falls, are in Appendix E.6.

5.5 When can pooling pay?

The results so far are consistent and small. This section asks whether their size is a property of the data or of our estimator, and shows that the two can be told apart with quantities the selector already computes on the fitting head (Section 3.4).

What gets selected, and how weakly. All five panels select $w < 1$: Carparts 0.90, Online Retail and M5 0.95, Auto 0.99, and RAF 0.997. Across three independent 5,000-series M5 samples the selected configuration is identical in structure, discount, and mixture order—(mixture, $w=0.95$, $K=6$) in every case.

The pooling axis behaves differently, and the honest summary is that it barely resolves. On three panels the spread between the best and worst pooling structure at the selected discount is under a tenth of a percent of validation SPL: 0.05% on Online Retail, 0.08% on Carparts, 0.02% on RAF. Only two panels separate at all—M5, where the learned mixture beats the global pool by 0.58%, and Auto, where the global pool beats every refinement by 0.31%. Announcing a winner on the other three would report selector noise. What the selector reliably does is choose a discount and avoid poor structural corners; it does not discover demand taxonomies, and we do not claim it does.

Leverage is an upper bound, not a prediction.

Table 5 reports the diagnostic. Read as Corollary 1 intends, it does not say which partition wins; it says how much any partition could win by. Two panels have almost no room on the size block: Auto and RAF retain median $\lambda^{(+)} \approx 0.92$ –0.94 even at their selected discounts, with no item below 0.1, so their sub-percent pooling differences are explained before any partition is fitted. The remaining three do have room. Carparts is extreme—median 0.224, with 22.5% of items essentially fully shrunk—and Online Retail, at 0.553 under the selected discount, sits squarely in the regime where Section 5.4 shows a supplied true partition earning more than two percent.

What binds is learnability, not leverage. Figure 3 places the real panels and the controlled panels on the same axes, and the gap between them is the

Table 5: **Credibility leverage under a single pool.** Median credibility weight per panel at no forgetting and at the selected discount, for both blocks of the hurdle model: $\lambda_i^{(o)} = n_i/(n_i + \phi_g)$ for occurrence and $\lambda_i^{(+)} = n_i^+/(n_i^+ + \kappa_g)$ for positive size, the latter being the quantity bounded by Proposition 2. A weight near one means the item’s own history dominates its posterior, bounding how far any shared prior can move it. “< 0.1” is the share of items whose size-block weight falls below 0.1. The final column is the external MAE change of a separately selected learned partition relative to the single pool; negative favors the single pool.

Panel	w^*	$\lambda^{(+)} \text{ (size)}$			$\lambda^{(o)} \text{ (occ.)}$		learned vs. pool
		$w=1$	w^*	< 0.1	$w=1$	w^*	
Carparts	0.90	0.802	0.224	22.5%	0.882	0.628	−4.8%
Online Retail	0.95	0.897	0.553	14.6%	0.969	0.895	−2.9%
M5	0.95	0.996	0.841	12.9%	0.998	0.931	−2.8%
Auto	0.99	0.944	0.940	0.0%	0.525	0.422	−3.6%
RAF	0.997	0.926	0.917	0.0%	0.085	0.106	−31.5%

section’s main result. Where the partition is correct by construction, the gain rises as leverage rises, reaching +2.2% near $\lambda^{(+)} \approx 0.5$. Where the partition must be learned, the gain is between −0.31% and +0.03% on four panels and +0.58% on the fifth, regardless of leverage. Online Retail is the decisive case: the selected discount moves it to $\lambda^{(+)} = 0.553$, inside the region where the controlled study says a good partition is worth more than two percent, and the mixture returns −0.05%. The occurrence block behaves the same way (Figure 3, bottom): RAF and Auto occupy the low- $\lambda^{(o)}$ end, where the shared prior supplies most of $\hat{\pi}_i$, and their refinement gains are still zero or negative.

This separates two explanations that aggregate results cannot. It is not that these panels lack exploitable cross-series structure—the credibility weights say the structure would be worth something if it were correct. It is that the mixture objective does not recover a partition worth shrinking toward. The binding constraint is learnability, and Corollary 1 bounds the prize rather than the shortfall.

High leverage is an amplifier with no sign attached. The clearest illustration is RAF, which combines the highest size-block leverage in the panel set ($\lambda^{(+)} = 0.917$) with the lowest occurrence leverage ($\lambda^{(o)} = 0.106$), and loses 31.5% of external MAE to a learned partition. The two facts are one fact. The point forecast is $\hat{y}_i = \hat{\pi}_i \exp(\hat{\mu}_i + \hat{\sigma}_i^2/2)$; when

Table 6: Online Retail distributional ablation. Each row is an independent hierarchical run; “auto” denotes its own internal discount selection. The raw predictive law is used and lower is better. Full point metrics are in Appendix Table 8.

Pooling	Forget	SPL	SPL ₉₀	AIW
global	off	0.8899	1.5306	10.51
taxonomy	off	0.8897	1.5299	10.56
learned	off	0.8905	1.5336	10.79
global	auto (.95)	<u>0.8849</u>	<u>1.5268</u>	9.25
taxonomy	auto (.95)	0.8846	1.5254	9.43
learned	auto (.95)	0.8869	1.5343	10.69

89% of $\hat{\pi}_i$ comes from the group prior, replacing the empirical content of the caveat attached to Proposition 1: persistent prior leverage guarantees that the choice of target keeps mattering, not that a particular target is good. Read together with the previous paragraph, low leverage caps the upside and high leverage uncaps the downside, and neither makes a learned partition worth using here.

Forgetting is an order of magnitude larger. Computed within a single selection surface under one definition—the M5 surface at its selected discount—refining the pooling structure is worth +0.58% of validation SPL, while discounting at the same structure is worth 8.95% for the global pool and 9.46% for the mixture. The asymmetry that motivates the paper survives at an order of magnitude, and it survives on the one panel where refinement is measurable at all.

What the ablation confirms. Table 6 is the external counterpart on Online Retail: the three pooling candidates differ by less than 0.1% in mean SPL without forgetting and by less than 0.3% at the selected discount, matching the internal picture. Discounting shows up elsewhere, narrowing the global and taxonomy 80% intervals by 11–12% while improving mean SPL by about half a percent; the full ablation (Appendix Table 8) shows it improves external MAE by 3.5–4.0% and slightly worsens RMSE. The cross-panel result in Table 2 should therefore be attributed to the shared hurdle-EB predictive law and to the temporal half of the selection, not to a learned-partition gain. That is not a concession made after the fact: the leverage diagnostic states the ceiling in advance, and Figure 3 shows we do not reach it.

6 Limitations

(1) The size law. The log-normal positive-size block keeps item-posterior updates analytic, but its tail is conservative on panels with heavy-tailed spikes. Official TweedieGP is better on Carparts and throughout that panel’s far tail, and conformal wrappers are better at q90 on M5. A Gamma or Tweedie size block inside the same hierarchy is the natural next instantiation, and Table 2 indicates where it would pay.

(2) Saturation and calibration. Raw intervals over-cover on the sparsest panels (Table 14), and on RAF the standard quantile levels saturate: the all-zero forecast attains a marginally lower mean SPL than every method. The far-tail levels ($q \geq 0.95$) resolve this (Appendix E.12), but CRPS and a cost-based criterion remain unrun.

(3) Assumptions behind the resolution condition. Corollary 1 uses a Gaussian working model for the item means and a χ^2 approximation for S_g^2 , accurate except for small groups combining strongly unequal counts with strongly unequal sampling variances, where it understates the collapse probability by up to 0.21. It is an estimator-specific estimability statement, not an information-theoretic impossibility result. Section 5.5 shows the practical consequence of that distinction: the bound is an upper envelope, and on these panels it is not the binding constraint.

(4) The learnability gap is demonstrated, not closed. Our strongest negative result—leverage exists and the learned partition does not exploit it—identifies a target rather than hitting it. Two extensions are directly implicated and neither is bounded by Corollary 1 in the same way. A soft-responsibility predictive mixture adds one matrix multiply and changes the *shape* of the predictive distribution rather than only the prior it shrinks toward, which is where Table 2 locates our remaining weakness. A partition objective that maximizes between-group separation subject to keeping $\rho_g \sqrt{m_g}$ above threshold would target learnability directly rather than likelihood. We report the gap because we can measure it, not because we have closed it.

(5) Baseline and inference scope. We do not run external hierarchical Bayesian implementations [2, 18, 20]; claims about them are limited to their published computational profiles. Appendix E.13 closes the inference-method half of this gap—a four-chain full-posterior Gibbs sampler matches plug-in EB within 0.1% on SPL—so the EB approximation is not the distributional-accuracy bottleneck, but the model-class half remains open. Our adaptive conformal comparison is one ACI design, not an exhaustive study of weighted or model-specific conformal

methods; given how much it changed the baseline (Section 5.3), the remaining designs deserve more attention than they have received in this literature. DeepState, Chronos-Bolt, and Deep Renewal Process comparisons use subset or proxy protocols and are reported only in Appendix E.11.

(6) Criterion and deployment alignment. Selection uses a single pinball-based score; the selector tracks whichever criterion it is given (Section E.5), but a cross-objective study including an inventory cost criterion is future work. The discount is selected under the fixed-origin protocol and reused in walk-forward: we keep the initialization discount and the online block update distinct and do not compound them, and we do not evaluate re-selection at every block. Appendix E records the remaining protocol details, including the reduced refit cadence for AutoARIMA, AutoTheta, and iETS on Online Retail.

7 Conclusion

Intermittent-demand forecasting requires two structural choices: how strongly to discount the past, and how to pool evidence across items. REMIX makes both by chronological validation inside a scalable sufficient-statistic empirical-Bayes family, applying one recency operator to item- and group-level statistics alike. Across five public panels it attains the lowest mean scaled pinball loss on four, with an official Tweedie–Gaussian-process specialist ahead on Carparts by half a percent, and it leads under strict walk-forward updating on both a long daily and a short monthly panel. These are distributional results obtained from a closed-form, CPU-only, bit-reproducible predictive law; point forecasts are competitive but not uniformly best.

The two choices turn out to be asymmetric, and the asymmetry is the paper’s finding. Discounting is a first-order decision: all five panels select it, and on the one panel where both effects are measurable it is worth an order of magnitude more than refining the pooling structure. Refinement, by contrast, improves external accuracy on none of the five panels and exceeds selector noise on only one.

Why it fails is the more useful result. Forgetting has two opposing effects on pooling—it keeps a shared prior’s leverage alive while eroding the information that would resolve fine group structure—and the resulting credibility weight bounds how much any partition could be worth. That bound is real but not binding. Three panels retain substantial room, and one of them, Online Retail, sits at the leverage where a supplied true partition earns more

than two percent in controlled study; its learned partition returns -0.05% . The constraint on cross-series pooling in these data is therefore learnability rather than leverage: the structure would pay if it were correct, and current mixture objectives do not find it. Separating those two explanations required the diagnostic, and the diagnostic costs nothing beyond the fit that produces it.

Two evaluation conventions actively obscure this picture, and both are inherited rather than incidental. Absolute-error metrics reward the all-zero forecast on three of five panels and reverse the ranking of a known-correct partition in simulation; they should not be the sole basis for comparing intermittent-demand forecasters. Static split-conformal wrappers, the field’s standard probabilistic baseline, understate what conformal prediction achieves here by roughly twenty percent relative to an adaptive construction—a margin several times larger than most method differences reported on these panels, including our own.

The practical value of REMIX is accordingly not a claim of universal accuracy but a cheap, deterministic predictive distribution together with an ex ante account of when temporal adaptation and finer pooling can matter. The most direct next steps follow from the gap the diagnostic exposes: partition objectives that optimize resolvable separation rather than marginal likelihood, predictive mixtures that alter distributional shape rather than only the shrinkage target, and decision-aware sequential selection.

Reproducibility Statement

All five datasets are public (UCI 352; Kaggle M5-Accuracy; the gluon-ts **intermittent-datasets** branch for Auto and RAF; Zenodo record 4656022 for Carparts). Conversion and preprocessing scripts are included in the released code. The full pipeline uses deterministic low-dimensional EB optimization and seed-free quantile-based EM initialization, so no seed policy is required for the forecaster itself. The stochastic elements are the seed-controlled M5 subsample and the $B=20$ bootstrap; M5 results are reported for seed 42 in the main tables and audited across three independently drawn samples, which select the same configuration in every case. An automated test asserts that two end-to-end runs are bit-identical. Every table maps to one experiment directory via a manifest file, and each run writes its selection diagnostics—chosen partition, discount, BIC curves, and credibility leverage—alongside its metrics. Hardware and timing methodology are spec-

ified in Appendix D. The complete experiment suite runs on one CPU workstation in under a day, dominated by the statistical baselines rather than REMIX. An anonymized code repository will accompany the submission.

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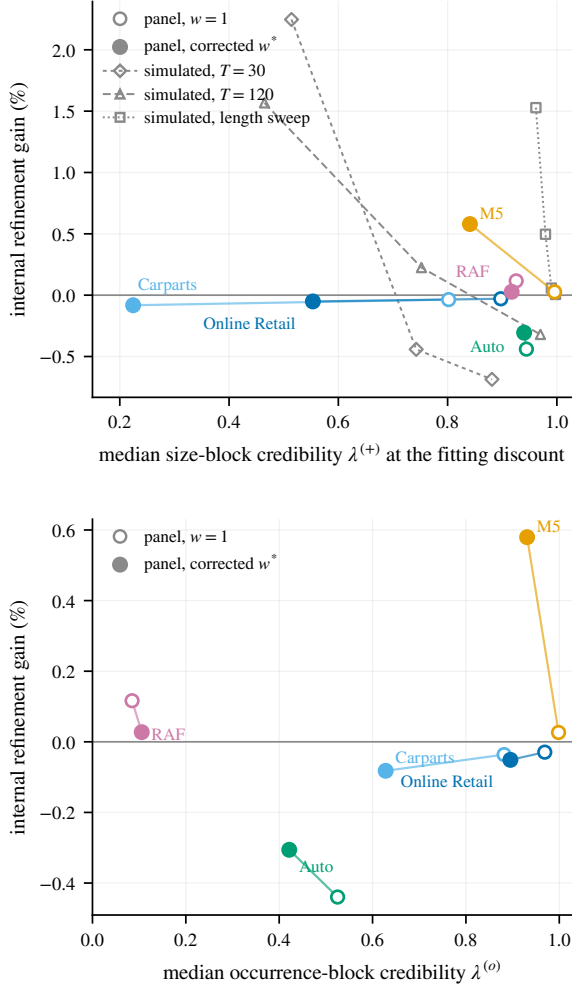


Figure 3: **The simulated curves are an upper envelope that the real panels do not reach.** Internal refinement gain (best refined structure versus the global pool, validation SPL) against median credibility weight, on the size block (top) and the occurrence block (bottom). Open markers are $w=1$, filled markers the selected discount, and the connecting segment is the movement forgetting induces. Grey curves are the controlled panels of Section 5.4, where the partition supplied is correct by construction. Every real panel lies far below the envelope at its own leverage—most visibly Online Retail, which the selected discount moves into the high-return region of the size block while its learned structure returns -0.05% .

A Notation

$y_{i,t}, z_{i,t}, \ell_{i,t}$	demand, occurrence indicator, log-size
T_i	initialization-window length of item i
$g(i), K$	pooling group of item i ; number of mixture components
α_g, β_g	Beta occurrence prior; $\phi_g = \alpha_g + \beta_g$ its strength
$\mu_{0,g}, \tau_g^2, \sigma_g^2$	size prior location, between/within-item variances
$w; b$	recency discount; forgetting strength $b = 1 - w$
$n_i(w), c_i(w)$	discounted observation / occurrence pseudo-counts
$\bar{p}_i(w)$	discounted empirical occurrence rate $c_i(w)/n_i(w)$
$\lambda_i(w)$	credibility weight $n_i(w)/(n_i(w) + \phi_g)$
$\hat{\pi}_i, \hat{\mu}_i, \hat{\sigma}_i^2$	posterior means of occurrence, log-size, variance
α_p	classical TSB occurrence smoothing constant ($= 1 - w$)
δ	occurrence drift rate (Proposition 4)
M, n	candidate count and validation size (Prop. 5)
B	bootstrap draws for hyperparameter averaging

B Supporting Results

This appendix records four connections that support the construction but are not the paper’s main theoretical contribution: the relation to TSB, a tracking bound under drift, finite-class selection, and classical credibility.

B.1 Relation to the TSB occurrence recursion

Write $p_t = z_{i,t}$ for one fixed item. Let $\hat{p}_T$ be classical TSB’s occurrence estimate with smoothing $\alpha_p = 1 - w$ and initialization $\hat{p}_0$.

Proposition 3 (TSB occurrence as a boundary case). For $w \in (0, 1)$: (i) $\lim_{(\alpha, \beta) \rightarrow (0, 0)} \hat{\pi}(w) = \bar{p}(w)$; (ii) $\hat{p}_T = (1 - w) \sum_{t=1}^T w^{T-t} p_t + w^T \hat{p}_0$ and $\bar{p}(w) = (\hat{p}_T - w^T \hat{p}_0)/(1 - w^T)$; (iii) hence $\bar{p}(w) = \hat{p}_T$ under the self-consistent initialization $\hat{p}_0 = \bar{p}(w)$, and $|\bar{p}(w) - \hat{p}_T| \leq w^T$ in general.

The equivalence concerns only the occurrence block. Classical TSB uses arithmetic smoothing for positive sizes, whereas REMIX discounts positive log-sizes under (1).

B.2 Tracking bound under occurrence drift

Proposition 4 (Tracking risk under drift). If $p_t \sim \text{Bern}(\pi_t)$ independently and $|\pi_t - \pi_T| \leq \delta(T - t)$, then $|\mathbb{E}\bar{p}(w) - \pi_T| \leq \delta w / [(1 - w)(1 - w^T)]$ and $\text{Var}\bar{p}(w) \leq \frac{1}{4}(1 - w)(1 + w^T) / [(1 + w)(1 - w^T)]$, with the corresponding bias-variance decomposition for the posterior mean.

Corollary 2. Balancing the two bounds gives $1 - w^* \asymp \delta^{2/3}$, worst-case MSE $O(\delta^{2/3})$, and $w^* \rightarrow 1$ as $\delta \rightarrow 0$.

This result motivates including $w = 1$ in the candidate grid; it does not claim that chronological validation estimates the bound-optimal w^* .

B.3 Finite candidate selection

Proposition 5 (Conditional finite-class bound). Let $R(c)$ be the expected validation risk of candidate c and $\bar{R}(c)$ its average over n validation items. If the item losses are independent conditional on the fitted candidates and lie in $[0, B]$, then the minimizer $\hat{c}$ over M candidates satisfies, with probability at least $1 - \eta$,

$$R(\hat{c}) \leq \min_c R(c) + B \sqrt{\frac{2 \log(2M/\eta)}{n}}.$$

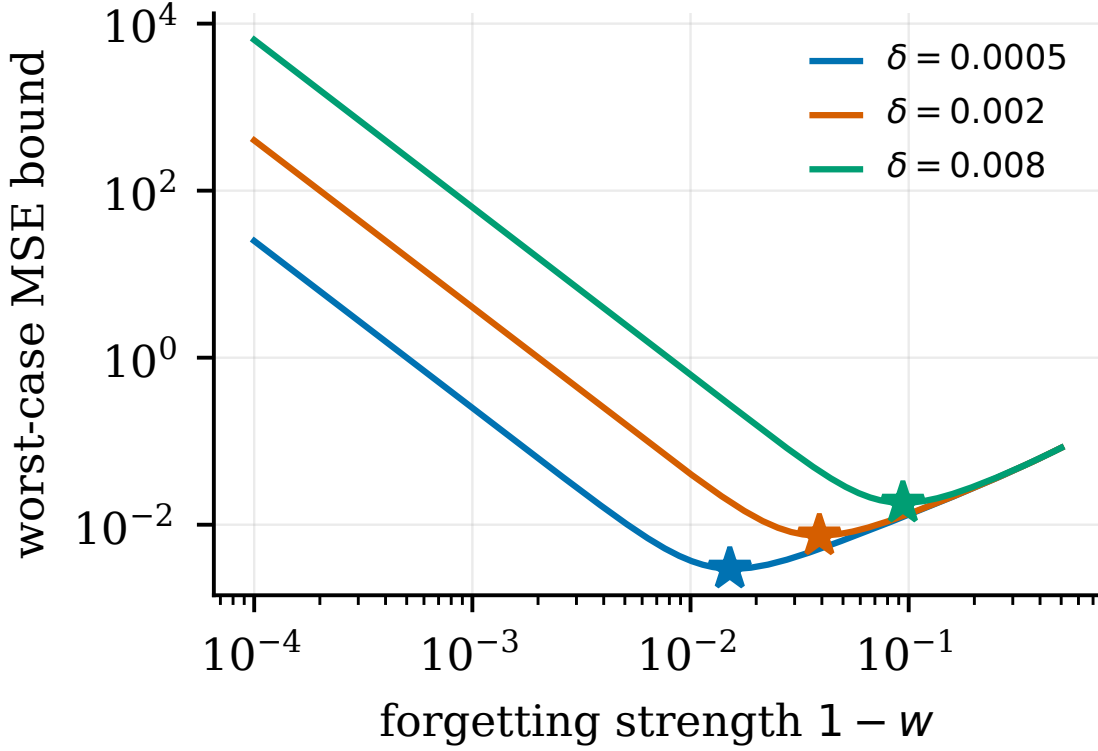


Figure 4: Bias–variance bounds in Proposition 4. Stars mark the minimizing forgetting strength; as drift δ increases, the optimum moves at the rate in Corollary 2.

The independent units are items, because observations and quantiles are averaged within item before selection. The released selector does not clip scaled pinball loss, so the bounded-loss assumption is not automatic and this proposition is not presented as a formal guarantee for the unclipped implementation. It records the standard finite-class behavior under an explicit clipping condition.

B.4 Classical credibility

Proposition 6 (Pooling reduces Bayes risk in the stationary model). Under the undiscounted Beta–Binomial occurrence model with known hyperparameters, the posterior mean is the Bayes estimator of π_i under squared loss and has strictly smaller Bayes risk than the per-item empirical rate whenever $\phi_g > 0$. Its optimal weight is the classical credibility factor $\lambda_i = n_i / (n_i + \phi_g)$ [1].

This is the Bühlmann–Straub result specialized to Bernoulli occurrences. It justifies pooling when the stationary hierarchical model is correctly specified; it does not assert that a misspecified or stale group prior helps under drift.

C Proofs

C.1 Proof of Proposition 3

(i) Immediate from (6) since $c(w), n(w)$ are finite. (ii) Induction: $\hat{p}_1 = (1 - w)p_1 + w\hat{p}_0$. If the closed form holds at $T - 1$, then $\hat{p}_T = (1 - w)p_T + w\hat{p}_{T-1} = (1 - w) \sum_{t=1}^T w^{T-t} p_t + w^T \hat{p}_0$. With $n(w) = (1 - w^T) / (1 - w)$,

$$\bar{p}(w) = \frac{(1 - w) \sum_{t=1}^T w^{T-t} p_t}{1 - w^T} = \frac{\hat{p}_T - w^T \hat{p}_0}{1 - w^T}.$$

(iii) Rearranging, $\bar{p}(w) - \hat{p}_T = w^T(\bar{p}(w) - \hat{p}_0)$, and $|\bar{p}(w) - \hat{p}_0| \leq 1$. $\square$

C.2 Proof of Proposition 1

$n(w) = (1 - w^T)/(1 - w) \leq 1/(1 - w)$ and $x \mapsto x/(x + \phi)$ is increasing, so $\lambda(w) \leq [1/(1 - w)]/[1/(1 - w) + \phi] = 1/(1 + (1 - w)\phi)$. $\square$

C.3 Proof of Proposition 4 and Corollary 2

With $k = T - t$ and $\sum_{k \geq 0} kw^k = w/(1 - w)^2$,

$$|\mathbb{E}\bar{p}(w) - \pi_T| = \left| \frac{\sum_t w^{T-t}(\pi_t - \pi_T)}{n(w)} \right| \leq \frac{\delta w/(1 - w)^2}{(1 - w^T)/(1 - w)} = \delta \frac{w}{(1 - w)(1 - w^T)}.$$

By independence and $\pi_t(1 - \pi_t) \leq \frac{1}{4}$,

$$\text{Var } \bar{p}(w) \leq \frac{\frac{1}{4}(1 - w^{2T})/(1 - w^2)}{[(1 - w^T)/(1 - w)]^2} = \frac{1}{4} \cdot \frac{(1 - w)(1 + w^T)}{(1 + w)(1 - w^T)}.$$

Decomposing $\hat{\pi} - \pi_T = \lambda(\bar{p} - \mathbb{E}\bar{p}) + \lambda(\mathbb{E}\bar{p} - \pi_T) + (1 - \lambda)(\mu_\pi - \pi_T)$, the first term is zero-mean and independent of the deterministic remainder, giving $\mathbb{E}(\hat{\pi} - \pi_T)^2 \leq [\lambda B(w) + (1 - \lambda)|\mu_\pi - \pi_T|]^2 + \lambda^2 \text{Var } \bar{p}(w)$ with $B(w)$ the bias bound. For the corollary set $b = 1 - w$ and let $T \rightarrow \infty$ (so $w^T \rightarrow 0$): the squared bias bound behaves as $\delta^2(1 - b)^2/b^2 \sim \delta^2/b^2$ and the variance bound becomes $\frac{1}{4}b/(2 - b)$, which is $b/8$ to leading order as $b \rightarrow 0$. Minimizing $f(b) = \delta^2/b^2 + b/8$ gives $f'(b) = -2\delta^2/b^3 + 1/8 = 0$, i.e., $b^* = (16\delta^2)^{1/3} = 2^{4/3}\delta^{2/3}$, and $f(b^*) = O(\delta^{2/3})$. $\square$

C.4 Proof of Proposition 5

For each candidate c , Hoeffding's inequality for independent scores in $[0, B]$ gives $\Pr(|\hat{R}(c) - R(c)| > t) \leq 2\exp(-2nt^2/B^2)$. A union bound over the M candidates makes $\sup_c |\hat{R}(c) - R(c)| \leq t$ hold with probability at least $1 - 2M\exp(-2nt^2/B^2)$. Setting the right side to η yields $t = B\sqrt{\log(2M/\eta)/(2n)}$. On this event, $R(\hat{c}) \leq \hat{R}(\hat{c}) + t \leq \hat{R}(c^*) + t \leq R(c^*) + 2t$ with $c^* = \arg \min_c R(c)$, which is the claim with constant $2t = B\sqrt{2\log(2M/\eta)/n}$. $\square$

C.5 Proof of Proposition 6

With known (α_g, β_g) and $s_i \mid \pi_i \sim \text{Bin}(n_i, \pi_i)$, $\pi_i \sim \text{Beta}(\alpha_g, \beta_g)$, the posterior mean $\hat{\pi}_i = \lambda_i \bar{p}_i + (1 - \lambda_i)\mu_\pi$ with $\lambda_i = n_i/(n_i + \phi_g)$ minimizes Bayes risk under squared loss among all estimators (tower property). Restricting to linear combinations $a\bar{p}_i + b$, the Bayes risk is $a^2 \mathbb{E}[\text{Var}(\bar{p}_i \mid \pi_i)] + (1 - a)^2 \text{Var}(\pi_i)$ at the optimal intercept, minimized at $a^* = \text{Var}(\pi_i)/(\text{Var}(\pi_i) + \mathbb{E}[\text{Var}(\bar{p}_i \mid \pi_i)]) = \lambda_i$, the Bühlmann credibility weight. The minimum is strictly below the risk of $a = 1$ (the empirical rate) whenever $\text{Var}(\pi_i) < \infty$ and $\phi_g > 0$. $\square$

C.6 Proof of Proposition 2 and Corollary 1

(i) $\lambda_i = n_i/(n_i + \kappa_g)$ is increasing in n_i , and $n_i \leq \sum_{k \geq 0} w^k = (1 - w)^{-1}$ for $w < 1$. Substituting the bound, $\lambda_i \leq \frac{(1 - w)^{-1}}{(1 - w)^{-1} + \kappa_g} = [1 + (1 - w)\kappa_g]^{-1}$, which is < 1 whenever $w < 1$ and $\kappa_g > 0$.

(ii) Write $\bar{y}_i = \mu_i + e_i$ with $\mu_i \sim N(\mu_g, \tau_g^2)$ and $e_i \sim N(0, s_i)$ independent, $s_i = \sigma_g^2/n_i$, so $\text{Var}(\bar{y}_i) = \tau_g^2 + s_i$. For the sample variance $S_g^2 = (m_g - 1)^{-1} \sum_{i \in g} (\bar{y}_i - \bar{\bar{y}})^2$, independence gives $\text{Var}(\bar{\bar{y}}) = m_g^{-2} \sum_i (\tau_g^2 + s_i)$ and hence

$$\mathbb{E}[S_g^2] = \frac{\sum_i (\tau_g^2 + s_i) - m_g \text{Var}(\bar{\bar{y}})}{m_g - 1} = \frac{(1 - m_g^{-1}) \sum_i (\tau_g^2 + s_i)}{m_g - 1} = \tau_g^2 + \bar{s}_g,$$

exactly, with no balance assumption. The estimator truncates precisely when $S_g^2 \leq \bar{s}_g$. Under the working model S_g^2 is approximately $(\tau_g^2 + \bar{s}_g)\chi_{m_g-1}^2/(m_g - 1)$, so $\text{Var}(S_g^2) \simeq \frac{2}{m_g-1}(\tau_g^2 + \bar{s}_g)^2$ and the standardized

distance from the truncation boundary is

$$\frac{\mathbb{E}[S_g^2] - \bar{s}_g}{\sqrt{\text{Var}(S_g^2)}} = \frac{\tau_g^2}{\sqrt{2/(m_g - 1)}(\tau_g^2 + \bar{s}_g)} = \sqrt{\frac{m_g - 1}{2}} \rho_g,$$

giving the stated normal approximation for $\Pr[S_g^2 \leq \bar{s}_g]$. The discount bound follows from $n_i \leq (1 - w)^{-1}$: $s_i = \sigma_g^2/n_i \geq (1 - w)\sigma_g^2$ for every i , hence $\bar{s}_g \geq (1 - w)\sigma_g^2$, and ρ_g is decreasing in $\bar{s}_g$. Finally $\hat{\tau}_g^2 = 0$ gives $\kappa_g = \infty$ and $\lambda_i = 0$.

Corollary. Requiring the collapse probability to stay below a fixed level is, by (ii), the condition $\rho_g \sqrt{(m_g - 1)/2} \geq c$ for a constant c of order one. Solving $\rho_g = \tau_g^2/(\tau_g^2 + \bar{s}_g) \geq c\sqrt{2/(m_g - 1)}$ for τ_g^2 gives $\tau_g^2 \geq \bar{s}_g c\sqrt{2}/(\sqrt{m_g - 1} - c\sqrt{2})$, i.e. the stated order $\bar{s}_g/(\sqrt{m_g - 1} - 1)$ up to the constant. Refining a partition decreases m_g and, by construction of the mixture objective, τ_g^2 , while decreasing w raises the floor $(1 - w)\sigma_g^2$ on $\bar{s}_g$. Each shrinks the left side or raises the right side. $\square$

The algebraic steps, the exact unbiasedness in (ii), and the monotonicity claims are verified with a computer algebra system in the released test suite. The χ^2 approximation in (ii) is checked against Monte Carlo simulation, agreeing to within 0.05 in probability across the regimes we use.

D Implementation details

EB fitting. Group Beta-Binomial hyperparameters maximize the (weighted) marginal likelihood by L-BFGS-B with positivity bounds; the size block uses pooled within-item variance for σ_g^2 and an intercept-only restricted marginal likelihood in τ_g^2 with profiled $\mu_{0,g}$; item-level process variances use the conjugate scaled-inv- χ^2 update with $\nu=20$. With discounting, every count is replaced by its weighted analogue (4); the Beta-Binomial marginal extends to real pseudo-counts via log-Gamma functions. The predictive law is zero-inflated log-normal with predictive variance $\hat{\sigma}_{\text{proc}}^2 + \hat{v}_\mu$. The released configuration averages quantiles over $B=20$ bootstrap resamples of the item-statistics rows. An optional location-scale calibration layer exists but is *off* in all headline results; the adaptive conformal comparison is reported as a separate baseline rather than used to modify REMIX after fitting.

Mixture EM. Deterministic quantile initialization over (occurrence rate, mean log-size); degenerate components (mass < 2) frozen to global fallback priors; tolerance 10^{-5} , max 60 iterations; $K \in \{1, \dots, 6\}$ by BIC with $5K + (K - 1)$ parameters.

Selection. Internal chronological split (first 80% / last 20%, min head 8); score = mean pinball over $q \in \{0.5, 0.75, 0.9\}$, per-series scaled by head mean absolute demand, averaged within item and then uniformly across items; grid over $w \in \{1, 0.999, 0.997, 0.995, 0.99, 0.98, 0.95, 0.90\}$ and partitions {global, taxonomy, mixture(BIC)}. Selected configurations per dataset are reported beneath Table 7; diagnostics (`remix_selection_diagnostics.csv`) accompany every run.

Environment and timing methodology. All experiments ran on one Windows 11 workstation (Intel Core Ultra 9 275HX, 32 GB RAM). Core determinism tests are bit-identical within this environment; cross-environment bit identity is not claimed. Timing is wall-clock and excludes data loading. Joint selection over the 24 candidates takes 51–243 seconds per panel, with one deterministic six- K EM/BIC search for the mixture on the fitting head. Final model-only point plus five-quantile prediction with $B = 20$ takes 6.2 seconds on Carparts, 38.0 on Online Retail, and 124.0 on M5, whose external span has 3.235 million observations. GPU is not used in REMIX.

Baselines. StatsForecast defaults; TSB $(\alpha_d, \alpha_p) = (0.5, 0.45)$; AutoARIMA/AutoTheta season 7 (daily) / 12 (monthly) for intervals. TSB-tuned: 5×5 grid $\alpha_d \in \{0.05, 0.1, 0.2, 0.35, 0.5\}$, $\alpha_p \in \{0.01, 0.05, 0.1, 0.25, 0.45\}$ (25 candidates, the same order as REMIX’s 24), scored on the identical 80/20 chronological split by per-series scaled MAE (so neither method is scored on its home criterion), refit on the full window. Selected (α_d, α_p) : Online Retail (0.10, 0.25); M5 (0.20, 0.01); Auto (0.20, 0.10); Carparts (0.50, 0.45), i.e. the textbook constants;

Table 7: Point forecasting across the five panels (lower is better; bold best and underline second among non-degenerate methods). Degenerate rows are shown as metric diagnostics and excluded from ranking.

Model	Online Retail		M5		Auto		Carparts		RAF	
	MAE	RMSSE	MAE	RMSSE	MAE	RMSSE	MAE	RMSSE	MAE	RMSSE
CrostonClassic	6.3294	5.1051	1.2345	2.3539	3.4056	1.1488	0.6792	1.0981	2.7315	1.6280
CrostonSBA	6.1953	5.0692	1.2076	2.3517	<u>3.3325</u>	<u>1.1451</u>	0.6628	1.0935	2.6520	1.6272
TSB	<u>5.5736</u>	4.8031	1.2672	2.3612	3.6587	1.2066	0.5396	1.0156	2.1660	1.6570
TSB-tuned	5.6318	4.7985	1.1932	<u>2.3137</u>	3.3917	1.1452	0.5396	1.0156	<u>2.2447</u>	1.6408
ADIDA	5.6860	<u>4.7967</u>	1.2274	2.3398	3.4079	1.1545	0.5556	1.0244	2.2977	1.6245
IMAPA	5.7000	4.7970	1.2317	2.3280	3.4143	1.1555	<u>0.5524</u>	1.0126	2.2850	1.6259
AutoARIMA	6.5494	4.8175	1.2283	2.3169	3.4417	1.1745	0.5822	<u>1.0105</u>	2.3153	1.6262
AutoTheta	8.2862	4.8550	1.2988	2.3143	3.7490	1.2215	0.5704	0.9966	2.3687	1.6399
Zero	4.7549	4.8089	1.2995	2.4629	4.1444	1.4454	0.3867	1.0949	1.1717	1.6345
Naive-last	6.0570	4.8324	1.3750	2.5450	4.3529	1.3818	0.5399	1.1408	1.8340	1.7644
REMIX (ours)	5.5325	4.7901	<u>1.2032</u>	2.2938	3.2145	1.1374	0.5726	1.0120	2.3874	1.6245

RAF (0.50, 0.25). Static conformal wrappers use a chronological 80/20 split and a *panel-pooled* signed-residual distribution; quantiles are zero-truncated and monotonized. The ACI variants retain that residual pool and adapt the requested level for every series-quantile pair with $\gamma = 0.05$. Tweedie-GLM is a per-series autoregressive GLM with log link, $p=1.5$, 14 lags (daily) / 6 (monthly), and $\ell_2=0.1$. TweedieGP uses the authors’ official implementation with Tweedie likelihood, median-demand scaling, RBF kernel, and 50,000 predictive samples; zero-only training series receive a zero fallback so that every series is scored. iETS uses `smooth::adam` with automatic occurrence modeling.

E Additional results

E.1 Point forecasting and horizon decomposition

Table 7 gives the complete point comparison retained for continuity with the intermittent-demand literature. Rankings exclude the deliberately degenerate Zero and Naive-last rows. REMIX takes six of ten MAE/RMSSE cells and is top two on seven, but the result is not uniform: classical methods retain several point-metric wins, and Zero attains the lowest MAE on Online Retail, Carparts, and RAF. This is why the main text uses point accuracy as a secondary diagnostic.

The selected (g, w) pairs are: Online Retail (global, 0.95), M5 (mixture, 0.95), Auto (global, 0.99), Carparts (global, 0.90), and RAF (mixture, 0.997). On three of the five panels the margins between pooling structures on the internal validation surface are under a tenth of a percent, so the structure component of these pairs should be read as Section 5.5 reads it: the selector avoids poor structural corners, and no winner is announced where the surface does not resolve one.

E.2 Selector diagnostics

The selection surface is locally flat near its optimum: on Online Retail the best seven candidates span all three pooling structures and lie within 0.51% of one another in validation SPL. Section 5.5 reports the credibility leverage and refinement gains that accompany each selected pair, and Figure 3 places them against the controlled panels. The selector is useful for choosing a discount and for avoiding poor structural corners; the winning pooling structure on a panel whose candidates lie within a tenth of a percent should not be read as a scientific taxonomy. Per-panel selection surfaces are released with the code.

E.3 Walk-forward point results and protocol

REMIX applies the selected discount once to the initialization sufficient statistics and then accumulates revealed blocks without compounding it. On Online Retail, ACI-ADIDA improves the static conformal

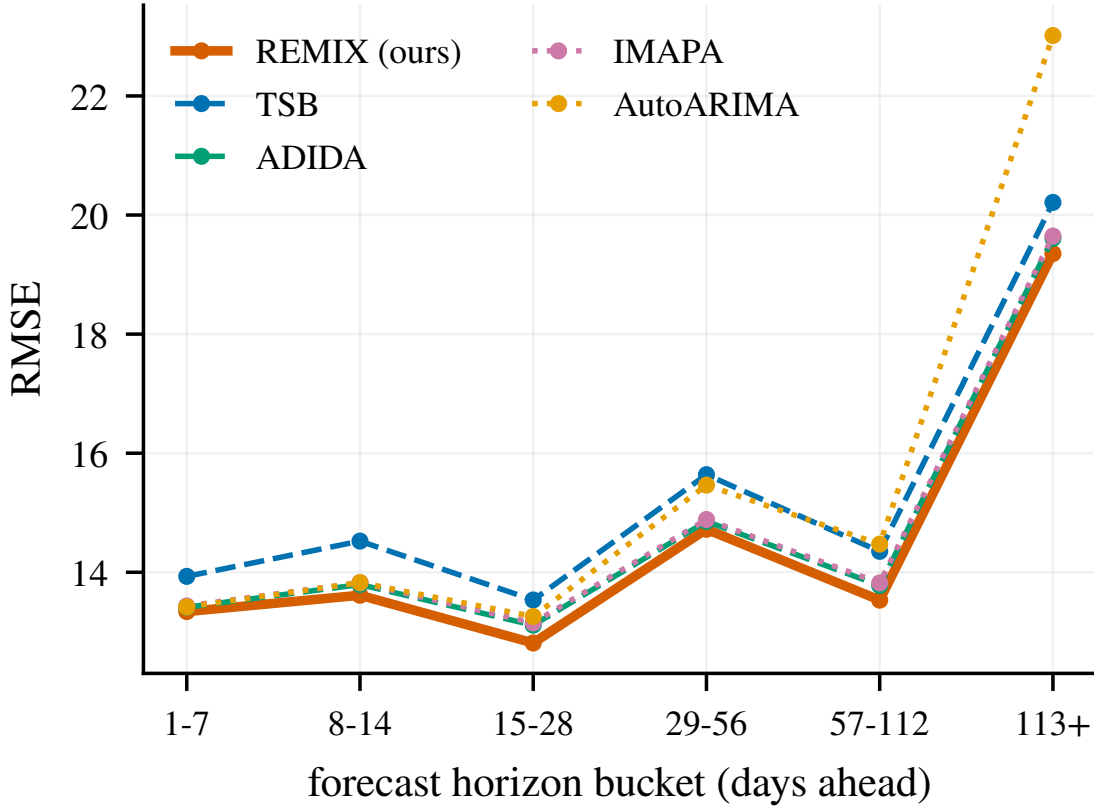


Figure 5: Online Retail RMSE by fixed-origin forecast-horizon bucket. REMIX is lowest in every bucket, so the aggregate result is not created by averaging only over distant horizons. MAE differences among the leading methods are within 1–2% by bucket.

wrapper substantially and wins q50 by a hair, but REMIX takes every other quantile, the mean, and q90 by a clear margin. On Carparts the first two quantiles saturate at zero and the discriminating gains occur from the median up. Re-running joint selection at every block is not evaluated.

E.4 Significance and interval diagnostics

Table 12 tests the headline metric directly: per-series mean SPL, REMIX against every comparator of Table 2, with paired t and Wilcoxon signed-rank tests Holm-adjusted within each panel and a comparison counted as decided only when both adjusted tests agree at the 5% level. REMIX is significantly better in 43 of the 48 panel-comparator pairs and significantly worse in none. The five undecided pairs are informative rather than embarrassing: on M5 the mean difference favors REMIX against all four conformal wrappers but the paired t does not reach significance after correction (the Wilcoxon strongly rejects symmetry, so the per-series differences are systematic but heavy-tailed), and on Carparts neither REMIX nor TweedieGP separates from the other (adjusted $p \geq 0.50$ both ways), consistent with the half-percent aggregate margin between them.

The paired MAE tests support a systematic improvement over the stationary hierarchy. Against TSB, tuned TSB, ADIDA, and IMAPA, mean MAE favors REMIX but the adjusted paired t -tests are not significant; the Wilcoxon tests reverse because those methods win on the typical series while REMIX reduces the difficult-series tail. On per-series RMSE, both adjusted tests favor REMIX against every comparator at $p < 0.05$. Raw intervals mildly over-cover, most strongly on Carparts.

Table 8: Full Online Retail ablation corresponding to Table 6. Each row is an independent run; “auto” denotes its own internal discount selection. TSB has no native predictive distribution. Lower is better.

Pooling	Forget	MAE	RMSE	RMSSE	SPL	SPL ₉₀	AIW
none (TSB)	fixed	5.5736	18.7272	4.8031	—	—	—
global	off	5.7623	17.6892	4.7851	0.8899	1.5306	10.51
taxonomy	off	5.7663	<u>17.6930</u>	<u>4.7875</u>	0.8897	1.5299	10.56
learned	off	5.8126	17.7064	4.7876	0.8905	1.5336	10.79
global	auto (.95)	5.5325	17.8386	4.7901	<u>0.8849</u>	<u>1.5268</u>	9.25
taxonomy	auto (.95)	5.5624	17.8421	4.7897	0.8846	1.5254	9.43
learned	auto (.95)	5.6925	17.8765	4.7902	0.8869	1.5343	10.69

Table 9: Online Retail point forecasting under strict seven-day walk-forward updates. The hierarchical rows differ in their initialization discount and pooling structure.

Model	MAE	RMSE	RMSSE
CrostonClassic	5.692	16.937	4.854
CrostonSBA	5.586	16.931	4.836
TSB	5.434	17.754	4.815
ADIDA	5.320	16.728	<u>4.637</u>
IMAPA	<u>5.332</u>	<u>16.745</u>	4.632
AutoARIMA	5.482	16.904	5.105
AutoTheta	5.394	16.836	4.735
Stationary hierarchy ($w = 1$)	5.586	17.260	4.725
Forgetting hierarchy (tax., .95)	5.411	17.059	4.682
REMIX (selected global, .95)	5.421	17.188	4.696

E.5 Selection objective and efficiency

The selector is not mathematically tied to pinball loss; it scores the criterion supplied on the chronological validation tail. We use item-averaged SPL because the target is a predictive distribution. The best candidates are locally close, and a full cross-objective regret matrix remains future work.

Joint selection takes 51–243 seconds across the five panels, excluding data loading, and covers 24 forecast candidates with one deterministic EM/BIC search for the mixture on the fitting head. Final estimation is analytic apart from low-dimensional group optimization; selection and point prediction are deterministic, while probabilistic bootstrap averaging uses the released fixed seed.

E.6 Controlled-study diagnostics

In the length sweep used to isolate credibility leverage, the generator holds $K = 2$, 500 items per group, occurrence probabilities 0.05 and 0.80, log-size centers three units apart, and $w = 1$. The true partition’s pinball advantage falls monotonically from 1.53% at $T = 12$ (median 4.5 positives per item) to 0.01% at $T = 400$ (median 154), with NLL, Brier, and log-size MSE following the same direction. Across the main grid, the RMSE of $\hat{\tau}_g^2$ approximately doubles as w falls from 1.00 to 0.90 ($0.037 \rightarrow 0.144$ at $T = 120$ and $0.074 \rightarrow 0.149$ at $T = 30$), directly measuring the resolution cost in Proposition 2.

The generator centers the groups symmetrically around a fixed grand mean so that increasing separation does not change total demand scale, and every arm is evaluated through the same hurdle-lognormal predictive representation under both likelihood-based proper scores and point metrics. A four-arm ladder—single pool, true labels with estimated hyperparameters, true labels with true hyperparameters, and the true conditional law—checks the implementation; the true conditional law is best on every metric in every cell.

Table 10: Full Online Retail probabilistic walk-forward results. SPL is reported at all five quantiles and on average; Cov@80 is marginal coverage and Cov⁺@80 conditions on positive demand. CP and ACI wrappers refit each block; AutoARIMA, AutoTheta, and iETS refit every four blocks; REMIX updates sufficient statistics each block.

Model	q10	q25	q50	q75	q90	Mean	Cov@80	Cov ⁺ @80
CP-Croston	0.341	0.778	1.364	1.745	2.632	1.372	0.793	0.479
CP-SBA	0.334	0.762	1.343	1.725	2.722	1.377	0.793	0.482
CP-TSB	0.451	0.899	1.289	1.385	2.842	1.373	0.795	0.468
CP-ADIDA	0.277	0.627	1.050	<u>1.319</u>	2.514	1.158	0.791	0.473
CP-IMAPA	0.292	0.660	1.084	<u>1.329</u>	2.522	1.177	0.791	0.471
AutoARIMA	0.259	0.572	1.197	1.763	1.522	1.063	0.933	0.762
AutoTheta	0.232	0.584	1.276	1.717	<u>1.483</u>	1.059	0.931	0.765
iETS	0.236	0.505	1.302	1.562	1.940	1.109	0.598	0.166
ACI-ADIDA	<u>0.186</u>	<u>0.468</u>	0.909	1.324	1.668	<u>0.911</u>	0.904	0.646
ACI-TSB	0.189	0.482	0.948	1.384	1.709	0.942	0.904	0.641
Zero (diagnostic)	0.183	0.458	0.916	1.374	1.649	0.916	0.740	0.000
REMIX	0.183	0.458	<u>0.910</u>	1.319	1.400	0.854	0.901	0.621

Table 11: Full Carparts monthly walk-forward SPL. Every statistical baseline refits each block; REMIX updates sufficient statistics. The zero row is excluded from ranking.

Model	q10	q25	q50	q75	q90	Mean
AutoARIMA	0.151	0.282	0.537	0.644	0.484	0.419
AutoTheta	0.120	0.253	0.523	0.658	<u>0.474</u>	0.406
CP-TSB	0.141	0.301	0.487	<u>0.513</u>	0.476	0.384
CP-ADIDA	<u>0.096</u>	<u>0.230</u>	<u>0.411</u>	0.529	0.488	0.351
CP-IMAPA	0.101	0.235	0.413	0.518	0.479	<u>0.349</u>
iETS	0.272	0.368	0.493	0.532	0.515	0.436
Zero (diagnostic)	0.075	0.187	0.374	0.560	0.672	0.374
REMIX	0.075	0.187	0.372	0.481	0.458	0.315

E.7 Released tables and artifacts

Complete per-dataset tables with RMSE, WRMSSE, all quantile levels, coverage, interval widths, fixed-corner ablations, and runtimes accompany the code release, together with the selection surfaces and per-run diagnostics behind every table in this paper.

E.8 Machine-readable significance output

The complete per-series results behind Table 13 and Table 12, including the RMSE tests and the Holm-adjusted families, accompany the code release as machine-readable CSV files.

E.9 MASE

For completeness we report MASE (per-series MAE scaled by in-sample naive MAE), the intermittent-demand literature’s traditional point metric. It inherits the absolute-error pathology quantified in Section 5.2: the all-zero forecast attains the best MASE on Online Retail (1.84 vs. TSB 2.08, REMIX 2.41), Carparts, and RAF, and zero-leaning methods (TSB, IMAPA) lead among real forecasters. On Auto, REMIX (0.889) is second only to CrostonSBA (0.883). Full table in the repository (`mase_table.csv`). We treat MASE as descriptive, not decision-relevant, for the same reason as MAE.

Table 12: Per-series mean SPL paired tests: REMIX versus every comparator on each panel, Holm-adjusted within panel (paired t and Wilcoxon signed-rank; the family includes TweedieGP on the monthly panels). Cells count comparators by outcome at the 5% level; named entries are the comparators for which REMIX is *not* significantly better.

Panel	Family	Sig. better	Sig. worse	Undecided
Online Retail	9	9	0	—
M5	9	5	0	CP wrappers (4)
Auto	10	10	0	—
Carparts	10	9	0	TweedieGP
RAF	10	10	0	—

Table 13: Paired per-series MAE tests on Online Retail: REMIX versus each alternative. Negative mean differences favor REMIX; p -values are Holm-adjusted.

vs.	Mean diff	p_t	p_{Wilcoxon}
Stationary hierarchy	−0.165	$< 10^{-29}$	$< 10^{-11}$
TSB	−0.040	0.532	$< 10^{-155}$
TSB-tuned	−0.090	0.086	$< 10^{-72}$
ADIDA	−0.036	0.359	$< 10^{-88}$
IMAPA	−0.049	0.281	$< 10^{-97}$
CrostonSBA	−0.872	$< 10^{-26}$	$< 10^{-59}$
AutoARIMA	−0.891	$< 10^{-14}$	$< 10^{-12}$
AutoTheta	−2.644	$< 10^{-81}$	$< 10^{-104}$

E.10 Walk-forward artifacts

The complete point and probabilistic results are in Tables 9 and 3; released block-level artifacts retain the seven-day boundaries and every per-block prediction.

E.11 Neural baselines

Table 15 retains the earlier neural comparisons. Deep Renewal Processes (DRP) [27] are architecturally close to our occurrence-size decomposition, but the legacy subsample numbers are indicative only: we rebuilt the Python 3.8/MXNet 1.7 environment and launched the original code path, yet three configurations diverged to NaN between epochs 7 and 11 and failed during prediction. DeepAR and an RNN under prediction-only chaining reach lower MAE in one case but worse RMSE and RMSSE at both horizons. Neural decoding and a constant sufficient-statistic forecast are not timing-equivalent, so these rows should not support a broad efficiency ratio.

Two newly executed reference points provide a cleaner, narrower check. DeepState—the published Amazon ISSM successor available in GluonTS, not a reproduction of Seeger et al. [20]—obtains mean SPL 1.0652 on a 1,000-series Online Retail subset and 0.4608 on full Carparts. Chronos-Bolt-small obtains 0.8448 on an Online Retail 500-series subset, versus 0.8342 for REMIX on exactly the same subset: a small 1.3% gap. Its Carparts result is only a reference point because the same-source Monash panel appears in the Chronos data repository and potential training contamination cannot be excluded. These subset results motivate an efficiency, determinism, and transparency claim, not a large foundation- model accuracy advantage.

E.12 Far-tail quantiles on the sparsest panels

Scaled pinball at $q \in \{0.95, 0.975, 0.99\}$ on the two panels where the standard levels discriminate weakly, under the identical fixed protocol (REMIX raw, $B=20$; iETS not re-run at these levels). The zero forecast’s scaled pinball is *linear* in q (its pinball at level q is $q \mathbb{E}[y/\text{scale}]$), so the far tail is exactly where its apparent advantage must collapse; the question the table answers is who inherits the lead.

Table 14: Raw REMIX interval quality across panels at nominal 80% coverage. No post-hoc calibration is used.

	OR	M5	Auto	Carparts	RAF
Coverage@80	0.892	0.846	0.907	0.931	0.921
AIW@80	9.25	2.81	9.21	1.47	0.94

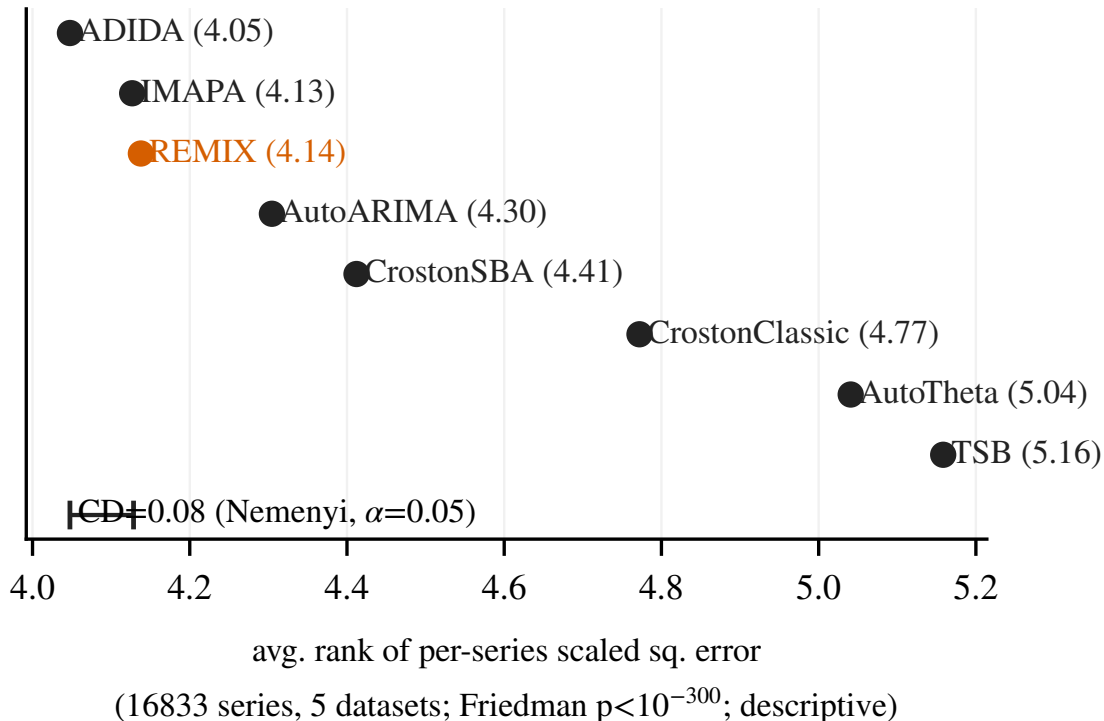


Figure 6: Average ranks of per-series scaled squared errors over the 16,833 series with complete saved row-level point predictions across the five panels. The critical-difference bar is shown only for scale: series, rather than datasets, are the blocks, so this is descriptive and not a valid Friedman–Nemenyi test.

Three readings follow. First, **the saturation breaks as expected**: on RAF the zero forecast, marginally ahead at $q \leq 0.9$, falls behind every intensity-bearing method from $q=0.95$ on, and on Carparts it is last by a factor of two. Second, **tail leadership is model-specific**: official TweedieGP wins all three Carparts cells, while REMIX is second in all three; on RAF, AutoARIMA wins q95, TweedieGP wins q97.5, and REMIX wins q99 by a near tie. Third, this table deliberately narrows the claim: REMIX’s most consistent advantage is aggregate SPL, not universal extreme-quantile dominance. The conformal wrappers still deteriorate in the RAF far tail, where their per-level calibration information is sparse.

E.13 Full-posterior inference on the same hierarchy

To separate the inference method from the model, we paired plug-in EB with a full-posterior sampler at each discount. Each pair uses the same single global pool, shared size-variance specification, discounted sufficient statistics, seed-controlled 500-series Online Retail subsample, evaluation span, and scoring code. The sampler uses Metropolis-within-Gibbs for the Beta hyperparameters, four deterministic chains of 4,000 sweeps, 2,000 burn-in iterations, and thinning by four. Both a stationary version ($w=1$) and a power-prior discounted version ($w=0.98$) are tested. The richer REMIX arm runs its joint selection separately on this subsample and

Table 15: Neural comparisons on Online Retail. Top: probabilistic (unscaled pinball, 500-series subsample, fixed protocol). Bottom: point metrics under prediction-only chaining at two horizons.

	q10	q25	q50	q75	q90	Time
DeepRenewal	0.358	0.895	1.790	2.684	2.941	~20 min
Hierarchical EB	0.387	0.914	1.772	2.390	2.186	1 s

	MAE		RMSE		RMSSE	
	$h=10$	$h=100$	$h=10$	$h=100$	$h=10$	$h=100$
RNN	3.257	3.132	11.938	12.538	0.875	8.197
DeepAR	6.245	7.288	17.221	18.623	7.151	9.986
Hierarchical EB	4.639	4.495	11.284	11.733	0.855	8.170

Model	Carparts			RAF		
	q95	q97.5	q99	q95	q97.5	q99
CP-Croston	0.413	0.290	0.168	0.488	0.537	0.526
CP-SBA	0.412	0.290	0.168	0.489	0.542	0.529
CP-TSB	0.374	0.263	0.160	0.506	0.601	0.552
CP-ADIDA	0.384	0.275	0.159	0.493	0.579	0.557
CP-IMAPA	0.378	0.270	0.156	0.493	0.581	0.564
AutoARIMA	0.385	0.332	0.301	0.460	0.395	0.356
AutoTheta	0.367	0.316	0.285	0.464	0.399	0.359
Tweedie-GLM	0.581	0.445	0.305	0.497	0.439	0.367
TweedieGP	0.355	0.222	0.128	0.462	0.352	<u>0.247</u>
Zero	0.710	0.729	0.740	0.507	0.520	0.529
REMIX	<u>0.365</u>	<u>0.251</u>	<u>0.144</u>	<u>0.461</u>	<u>0.364</u>	0.246

is reported only as context, not as an inference-method contrast.

	SPL mean	SPL q90	MAE	RMSSE	sec.
REMIX (incl. selection)	0.7942	1.3682	5.6259	4.6712	86.4
EB global, $w=1$	0.7919	1.3588	5.5504	4.6684	0.2
Gibbs global, $w=1$	0.7919	1.3590	5.5890	4.6696	2.7
EB global, $w=0.98$	0.7935	1.3658	5.4774	4.6694	0.2
Gibbs global, $w=0.98$	0.7937	1.3665	5.5549	4.6734	2.7

Two readings. First, *the empirical-Bayes approximation is not a distributional-accuracy bottleneck*: paired full-posterior inference changes mean SPL by 0.007%/0.019%, q90 SPL by 0.018%/0.048%, and RMSSE by 0.025%/0.085% at $w = 1/0.98$. MAE is somewhat more sensitive but still changes only 0.70%/1.42%. Second, forgetting helps MAE under both inference methods, so that finding is not an artifact of the EB plug-in. The maximum split- $\hat{R}$ across $(\alpha, \beta, \mu_0, \tau^2, \sigma^2)$ is 1.0021, and the minimum effective sample size is 819 of 2,000 retained draws. What this experiment does *not* measure: the richer external state-space samplers [2, 20] remain unrun (Limitations, item 6); this conjugate single-pool Gibbs is far cheaper than those models and is a control for the inference method, not a stand-in for them. The plug-in EB path retains bit-reproducibility and requires no convergence diagnostics, but at this scale raw speed does not separate the two.

E.14 Datasets

Online Retail: UCI 352; returns/zero-price removed, 99.5% quantity cap, daily aggregation densified with zeros. M5: Kaggle M5-Accuracy; seed-controlled 5,000-series sample. Both M5 protocols use a two-thirds initialization split; the learned mixture is selected at $w = 0.95$ on all three audited seeds. Auto/RAF: gluon-ts `intermittent-datasets` branch. Carparts: Zenodo 4656022, series without missing values and with at least

one zero (2,509). Monthly panels use the last- h holdout of Damato et al. [\[6\]](#).